Bangladesh University of Engineering and Technology

DEPARTMENT OF CIVIL ENGINEERING

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Course Title: Capstone Project

Feasibility Study Report

Development of a Multi-storied Shopping Complex at Mohammadpur Krishi Market

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Ministry of Commerce
Government of the People's Republic of Bangladesh

Feasibility Study for Development of a Multi-storied Shopping Complex at Mohammadpur Krishi Market.

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Project Feasibility Study Report

Section 1: Basic Information

1.	Name of the Project	:	Development of a Multi-storied Shopping Complex at Mohammadpur Krishi Market.
2.	(a) Sponsoring Ministry/Division (b) Implementing Agency	: :	Ministry of Commerce. Dhaka North City Corporation(DNCC).
3.	Project Objectives (Project to be taken based on the study)	:	• Enhance Market Infrastructure • Boost Local Economy • Promote Sustainability • Improve Accessibility and Safety • Improve Public Health and Safety • Collaborative Development
4.	Estimated Project Cost (Taka in Crore)	:	138 crore
5.	Sector & Sub-Sector	:	Commerce
6.	Project Category (Based on Environment Conservation Rules 1997)	:	Orange
7.	Project Geographic Location (a) Countrywide (b) Division (c) District (d) Upazila (e) Others (City Corporation/Pourashva)		 Bangladesh Dhaka Dhaka Dhaka North City Corporation
8.	Project Duration	:	2 years

Section 2: Introduction

2.1 Project Background

The Mohammadpur Krishi Market, situated in Dhaka North City Corporation (DNCC) Ward 29, has long served as a vital hub for the local community, providing essential services and economic opportunities. However, a devastating fire incident has left a significant portion of the market in ruins, creating a pressing need for redevelopment. The total area of the market is approximately 217,800 square feet (5 acres), with our project specifically focusing on a portion measuring around 1.11 hectares (2.74 acres). Currently, the site still has debris and remnants of the destroyed structures, posing both a challenge and an opportunity for revitalization.

The proposed redevelopment aims to transform this area into a multi-functional complex that addresses both community and commercial needs. The new building will feature a prayer space, a shopping mall, and a revitalized vegetable market. This redevelopment is not just about reconstructing what was lost but also about enhancing the functionality and appeal of the market to better serve the needs of the local population.

2.2 Objectives of the Feasibility Study

The objective of the feasibility study is to assess the overall viability of the Mohammadpur Krishi Market redevelopment project, considering alternatives, risks, market demand, financial sustainability, technical aspects, legal compliance, operational practicality, scheduling, and environmental and social impacts.

The redevelopment of a 1.11-hectare (2.74-acre) portion of the Mohammadpur Krishi Market in Dhaka aims to transform the existing market space into a modern, multi-functional complex. This feasibility study serves as the foundation for evaluating the technical, financial, environmental, and social dimensions of the project to ensure informed decision-making and project success.

Technical Feasibility Analysis:

This section evaluates the technical feasibility of the redevelopment project. It examines the current condition of the market's infrastructure, the proposed enhancements, and the incorporation of modern design elements. Special attention is given to the integration of resilient design standards, ensuring that fire safety, earthquake resistance, and disaster risk

reduction are embedded into the project's architectural and engineering framework. The feasibility study will also assess the capacity of the implementing agency, and the realistic scope of work required to complete the project within the projected timeline.

Economic and Financial Assessment:

The study evaluates the project's financial feasibility by examining the projected costs associated with redevelopment, including the construction of new market spaces, a shopping mall, a prayer area, and other commercial spaces. It analyzes the potential sources of funding, budget allocation, and the projected return on investment (ROI). Cost-benefit analyses are conducted to determine the economic impact of the project on local vendors and businesses. Financial sustainability is also a focus, with an emphasis on long-term operational costs and potential revenue generation from the new market facilities.

Environmental and Social Impact:

The environmental and social aspects of the project are examined in detail. The study assesses the environmental impact of the new development, including its alignment with eco-friendly design and sustainable practices. It explores how the project will affect local ecosystems, waste management, and resource consumption. From a social perspective, the study evaluates how the new market will serve the community, enhance the livelihoods of local vendors, and improve public accessibility. The goal is to ensure that the redevelopment contributes positively to the social and economic fabric of the area.

Vulnerability and Risk Reduction Analysis:

Risk assessment is a key component, focusing on identifying potential hazards such as flooding, fire, and other disasters. The study will incorporate strategies for mitigating these risks through resilient urban planning and disaster preparedness measures. Climate change adaptation, flood management, and emergency response systems are critical factors in ensuring the market's long-term resilience and operational continuity.

2.3 Approach and Methodology of the Feasibility Study

The initial step in conducting the feasibility study involves defining the project's scope to clearly identify its objectives and deliverables. After gathering all necessary data, a comprehensive market analysis is undertaken to assess competition and demand. The financial analysis projects potential revenues, estimates costs, and calculates key financial indicators, while the technical evaluation focuses on detailing the project's technological requirements. A risk assessment is carried out to identify potential challenges, and legal and regulatory compliance ensures that the project adheres to all relevant laws and standards. Operational feasibility is then evaluated to determine how well the project integrates with existing processes, followed by the development of realistic schedules and timelines. The environmental and social impact is examined to assess sustainability and social responsibility. Stakeholder analysis is thoroughly conducted, and all potential alternatives are reviewed. Finally, the entire study is documented and presented in detail.

2.4 Organization of the Feasibility Study

This report on the feasibility study is structured into 12 sections:

- **Section 1** introduces the project and provides fundamental information.
- **Section 2** outlines the project's background, objectives, and the approach and methodology applied in the feasibility study.
- **Section 3** focuses on the demand analysis, which includes a problem statement, the relevance of the project, possible interventions, project stakeholders, and a demand and SWOT analysis.
- **Section 4** presents the preliminary technical analysis and design, along with output plans, initial cost estimates, and the implementation schedule.
- **Section 5** addresses environmental sustainability, climate resilience, and disaster risk assessments.
- **Section 6** conducts a cost-benefit analysis, encompassing both economic and financial aspects.
- **Section 7** covers the human resources and administrative support required for the project.
- **Section 8** delves into the institutional and legal considerations of the project.

- **Section 9** assesses the project's risks and sensitivities.
- **Section 10** examines alternative options from structural and geotechnical perspectives for the proposed development.
- **Section 11** provides practical recommendations for the project's successful execution.
- **Section 12** contains the results of the demand analysis, a detailed Traffic Impact Assessment, preliminary designs, and cost estimates.

Section 3: Market/Demand Analysis

The Mohammadpur Krishi Market has long been an important commercial hub for the local community, but it currently lacks the infrastructure needed to meet the evolving demands of the area. The absence of a modern market, along with limited facilities and services, hinders economic growth and reduces the market's potential to serve the community effectively. Redeveloping this portion of the market into a multi-functional complex is crucial to addressing these shortcomings and revitalizing the local economy. By introducing a prayer space, shopping mall, improved parking, and a revitalized vegetable market, this project aims to modernize the area and support both commercial and community needs.

3.1 Problem Statement

The current state of the Mohammadpur Krishi Market is inadequate for serving the needs of the growing local population and does not provide the necessary infrastructure for vendors, shoppers, or community members. Key issues include:

I. **Absence of a Modern Market Space**

- The existing market lacks a well-organized, modernized structure, which limits the efficiency of vendors and reduces the quality of the shopping experience for customers. The absence of essential facilities affects the market's ability to attract a larger consumer base.

II. **Inadequate Parking Facilities**

- The market suffers from a lack of dedicated parking spaces, leading to traffic congestion, disorganized parking, and frustration for both shoppers and vendors. This

creates an unattractive environment for visitors, negatively impacting business operations.

III. **Lack of a Shopping Mall**

- The market does not feature a modern shopping mall that could house diverse retail stores, limiting the variety of goods and services available to consumers. This restricts the market's ability to compete with other commercial centers in the area and stifles local economic growth.

IV. **Limited Clothing and Specialty Shops**

- Currently, the market lacks a variety of shops offering clothing, accessories, and other specialty items. This absence reduces its appeal to a wider customer demographic and limits the potential for expanding local business opportunities.

V. **No Designated Prayer Space**

- The market does not include a designated prayer area, which is a key requirement for many visitors and vendors. The absence of such a space diminishes the market’s inclusivity and convenience, especially for religiously observant community members.

VI. **Limited Employment Opportunities**

- The underdeveloped state of the market infrastructure results in fewer job opportunities for local residents. The absence of a modernized market space limits the potential for new vendors, employees, and support staff, thereby hindering economic progress in the area.

VII. **Lack of Adequate Transportation and Accessibility**

- Poor transportation planning and inadequate accessibility make it difficult for customers and suppliers to reach the market efficiently. This affects business operations, limits customer footfall, and discourages vendors from expanding their services.

VIII. **No Recreational or Community Space**

- The market currently lacks any recreational or communal areas that could attract families and make the market a more inclusive and appealing destination. The absence of such spaces reduces the market's potential to become a central hub for the community, where people can gather not only for shopping but for social and leisure activities.

3.2 Relevance of the Project Idea

The redevelopment of the Mohammadpur Krishi Market is a critical initiative aimed at restoring and revitalizing a crucial commercial hub that has long served the local community. This project aligns with both national and global development agendas by addressing economic growth, infrastructure development, employment generation, and sustainable urban planning. By transforming a portion of the market into a multi-functional complex, including a prayer space, shopping mall, and revitalized vegetable market, the project will create a vibrant, inclusive environment that supports the community's socio-economic needs.

3.2.1 Project Goals

Economic Goals

- **Boost Local Economy:** Revitalize the local market to stimulate economic activity, promote business growth, and attract investments.
- **Create Employment:** Generate new jobs in retail, service, and maintenance sectors, contributing to local income generation and economic stability.
- **Support Entrepreneurship:** Provide affordable space and support for small business owners and entrepreneurs, encouraging local enterprise development.

Social Goals

- **Foster Community Engagement:** Develop a market that serves not only commercial but also social functions, offering communal spaces for leisure and religious activities.
- **Improve Quality of Life:** Provide better facilities and a safer, more organized market environment that enhances the daily lives of vendors and shoppers.

- **Promote Social Inclusion:** Ensure that the market caters to the diverse needs of the local population, including low-income families and women, by offering accessible and affordable services.

Infrastructure Goals

- **Upgrade Market Facilities:** Construct a multi-functional complex that incorporates modern amenities such as clean sanitation, organized vendor stalls, and sustainable building designs.
- **Enhance Mobility and Parking:** Improve market accessibility through the development of efficient parking spaces and well-planned access routes.
- **Incorporate Smart Infrastructure:** Integrate modern technology for better market management, including smart waste disposal, energy-saving systems, and digital payment solutions.

Environmental and Sustainability Goals

- **Implement Green Infrastructure:** Integrate eco-friendly practices such as solar energy, rainwater harvesting, and waste management systems into the market's design.
- **Reduce Environmental Footprint:** Promote sustainable market operations that reduce pollution, waste, and resource consumption, aligning with SDG targets on sustainable cities and communities.

3.2.2 Project Outcomes and Impact

The redevelopment project will have far-reaching effects on the community and the local economy. Key expected outcomes include:

- **Modernized Market Infrastructure**
  Outcome: A modern, well-structured market that enhances vendor efficiency and improves the shopping experience for customers.
  Impact: Increased customer footfall, improved vendor operations, and more organized commercial activities, leading to a stronger local economy.
- **Improved Accessibility and Parking Facilities**
  Outcome: The introduction of dedicated parking facilities will ease traffic congestion and provide convenient access for both shoppers and vendors.

Impact: Reduced traffic bottlenecks around the market area, enhanced customer convenience, and better overall market accessibility.

- **Diversified Retail Options**
  Outcome: The creation of a shopping mall with various retail shops, including clothing, specialty stores, and essential services.
  Impact: Increased variety of goods and services available to the local population, encouraging economic activity and drawing more visitors to the area.
- **Enhanced Employment Opportunities**
  Outcome: Job creation for local residents, including roles for vendors, service staff, and maintenance workers, generated by the expanded commercial space.
  Impact: Increased employment and income for local families, contributing to poverty alleviation and improved living standards.
- **Community-Centric Amenities**
  Outcome: The inclusion of a prayer space and communal areas will cater to the spiritual and social needs of the market's diverse population.
  Impact: A more inclusive market environment that fosters community engagement, making the market not just a commercial but also a social and cultural hub.
- **Sustainable Market Operations**
  Outcome: The redevelopment project will incorporate sustainable practices, such as energy-efficient buildings, waste management systems, and improved sanitation.
  Impact: A cleaner, eco-friendly market that supports long-term urban sustainability goals while providing a healthier environment for vendors and visitors.
- **Increased Transportation and Logistical Efficiency**
  Outcome: Better transportation planning and improved access routes for goods and customers.
  Impact: Enhanced efficiency in the movement of goods to and from the market, reducing operational costs for vendors and improving the overall supply chain.

3.3 Proposed Project Interventions

The redevelopment of the Mohammadpur Krishi Market requires a diverse approach to ensure that it addresses both immediate and long-term needs of the community. Key interventions include:

➢ **Modern Market Design:** Establish a well-organized, multi-functional complex that includes a prayer space, shopping mall, and a revitalized vegetable market.
➢ **Parking Solutions:** Develop a dedicated parking area to reduce congestion and improve access to the market.
➢ **Diverse Retail Opportunities:** Introduce a shopping mall that accommodates clothing, specialty shops, and essential services, providing greater variety and enhancing the market's appeal.
➢ **Community-Centered Spaces:** Incorporate prayer areas and recreational spaces to make the market a central hub for both commerce and community engagement.
➢ **Sustainable Infrastructure:** Implement energy-efficient technologies and eco-friendly practices to minimize the environmental impact of market operations.
➢ **Enhanced Transportation:** Improve transportation routes and access for vendors and customers, ensuring seamless logistics and reducing congestion.

3.4 Stakeholders:

In the case of the redevelopment of the Mohammadpur Krishi Market, the key stakeholders involved in the project interventions include:

Local Community and Businesses:

Residents and local business owners who rely on the market for their daily needs and livelihood. The redevelopment will impact their access to essential goods, services, and job opportunities. It will also influence the community’s economy, creating more jobs and attracting more customers.

Government Authorities and Agencies:

Dhaka North City Corporation (DNCC) plays a crucial role in overseeing the project. Additionally, other local government bodies, urban planning authorities, and regulatory agencies will ensure compliance with safety standards, land use, and building codes, and issue necessary permits.

Market Vendors and Traders:

The existing vendors and traders in the Krishi Market who will be affected by the redevelopment. They are key stakeholders whose relocation, rehousing, and participation in the new market structure are critical for the project's success.

Construction and Development Teams:

Architects, engineers, contractors, and workers who are directly responsible for the design and construction of the multi-functional complex. Their expertise ensures the project meets timelines, budget requirements, and safety standards. All civil engineers, MEP engineers and experts that involved in the project are also stakeholders.

Environmental and Civic Organizations:

These organizations assess the environmental sustainability and social impact of the redevelopment. They ensure that the project follows eco-friendly practices and promotes the well-being of the surrounding community through responsible construction methods and energy-efficient designs.

Transportation and Infrastructure Departments:

These agencies are tasked with improving access to the new complex, addressing transportation challenges, and managing infrastructure upgrades, such as roads, parking facilities, and public transport to accommodate increased traffic and footfall in the area.

Financial and Investment Partners:

Banks, investors, and financial advisors who are involved in securing funding for the project, managing the budget, and ensuring the long-term profitability of the redeveloped market. Their role is vital for financial planning, risk management, and monitoring the economic returns of the project.

3.5 Demand Analysis

To assess and forecast the demand for the reconstructed Mohammadpur Krishi Market, a demand survey was conducted. This survey targeted both sellers and customers within the market area. The objective was to gather insights on current needs, preferences, and expectations from those directly impacted by the market's redevelopment. The survey included local vendors, current market goers, and residents of the surrounding community to ensure a broad understanding of market demand and to inform the design and functionality of the new multi-functional complex.

Key Takeaways from the survey

- **Storage and Management**: Approximately 60% of sellers indicated a need for improved storage facilities, and another 60% emphasized the necessity for better trash management, which was previously inadequate.
- **Pricing Concerns**: About 75% of sellers reported that the prices they received were low compared to other markets. Lot of buyers also thought the pricing was reasonable.
- **Market Frequency**: Around 32% of marketgoers visited the Mohammadpur Krishi Market regularly, indicating a steady customer base.
- **Customer Comfort**: Approximately 60% of customers expressed a desire for improved seating areas within the market to enhance their shopping experience.
- **Location Feedback**: About 46% of customers rated the market's location as very good, indicating favorable opinions on its accessibility and convenience.

Constraints

The redevelopment of Mohammadpur Krishi Market faces several constraints that could impact its success:

Financial Constraints: Limited budget can restrict the quality and scope of the project, while unexpected cost increases could strain finances. Securing long-term funding for maintenance and upgrades remains a challenge.

Regulatory Constraints: Navigating complex approval processes and ensuring compliance with building codes and environmental regulations may delay the project and add to its complexity.

Operational Constraints: Managing multiple stakeholders and resources efficiently can be challenging, potentially affecting project timelines and quality. Ensuring ongoing maintenance and timely upgrades may face delays due to financial or bureaucratic issues.

Environmental Constraints: Implementing sustainable practices and managing environmental impacts during construction may be constrained by budget and technical feasibility.

Technological Constraints: Integrating advanced technologies into the redevelopment could be limited by financial constraints or compatibility issues, while logistical challenges during construction might arise due to access limitations.

Stakeholder Constraints: Balancing the expectations of local businesses, vendors, and the community with project goals can be difficult, and political influences may impact funding and project priorities.

Timing Constraints: Delays in approvals, construction, or procurement processes can affect the overall project timeline, potentially impacting its alignment with market operations.

3.6 SWOT Analysis

Strengths

1. **Enhanced Community Hub**: Reconstructing the Krishi Market with additional facilities will strengthen its role as a central, multifunctional community space, fostering greater local engagement and convenience.
2. **Increased Economic Activity**: The new market design, including a shopping mall and expanded vendor spaces, will boost local businesses and create new economic opportunities.
3. **Improved Infrastructure**: The inclusion of modern amenities such as parking facilities and a prayer space addresses current deficiencies and enhances overall market functionality.

4. **Environmental and Social Impact**: The redevelopment can incorporate sustainable practices and offer social benefits by providing a revitalized space for community interaction and economic development.
5. **Government and Community Support**: The project aligns with urban renewal goals and can garner support from both local authorities and community members, potentially easing approvals and funding.

Weaknesses

1. **Funding Constraints**: Securing sufficient funding for the expanded scope, including advanced infrastructure and amenities, may be challenging and subject to budget fluctuations.
2. **Construction Delays**: The scale of redevelopment could face delays due to unforeseen construction issues, regulatory approvals, or logistical complications.
3. **Bureaucratic Hurdles**: Navigating local government regulations and obtaining necessary permits can be time-consuming and impact project timelines.
4. **Community Disruption**: During construction, local businesses and daily activities may experience disruptions, affecting community support and market operations.
5. **Integration Challenges**: Ensuring seamless integration of new facilities with the existing market infrastructure may present design and operational challenges.

Opportunities

1. **Revitalization of the Area**: The project offers an opportunity to revitalize a key market area, improving both its economic viability and community appeal.
2. **Enhanced Local Amenities**: Adding features such as a modern shopping mall and dedicated prayer space can attract more visitors and serve diverse community needs.
3. **Job Creation**: The redevelopment will create jobs in construction, retail, and management, benefiting the local economy and providing employment opportunities.
4. **Increased Property Values**: The upgraded facilities and improved infrastructure may increase property values in the surrounding area, contributing to overall urban development.
5. **Potential for Future Expansion**: The new market design may lay the groundwork for future expansion or additional services based on evolving community needs.

Threats

1. **Economic Instability**: Economic downturns or shifts in market conditions could impact funding availability and project feasibility.
2. **Regulatory Changes**: New or changing regulations related to construction, environmental standards, or urban planning could affect project implementation.
3. **Public Resistance**: Community opposition or dissatisfaction with the redevelopment process could arise, potentially leading to delays or additional challenges.
4. **Construction Risks**: Unexpected issues during construction, such as safety hazards or logistical problems, could impact project timelines and costs.
5. **Maintenance**: Ensuring long-term maintenance and operation of the new facilities may be challenging, especially if budget constraints or other issues arise.

Section 4: Technical/Technological & Engineering analysis

4.1 Location:

The proposed project site is located at Mohammadpur Krishi Market, under the jurisdiction of Dhaka North City Corporation (DNCC) Ward 29. The boundaries of the project area are defined by narrow roads on the north and west sides. The total land area of the market is approximately 217,800 square feet (5 acres). Our project area is around 1.11 Hectare (2.74 Acre). Currently, the site has various debris and broken parts of structure from the fire incident. This project aims to reconstruct on part of that space which will be about 300’ in length and 250’ in width.

Coordinates of the location: **23.766427228952473, 90.3604400218135**



Figure: Location of project on Google Maps



Figure: Satellite view of project location

4.2 Technical Design:

The proposed project includes kacha bazar, shopping complex, mosque and additional place for other activities. The constructed building will be three storied (G+2) with 3 basements for parking. The building will have necessary structural integrity for future expansion up to 5 additional floors (G+7). There will be better parking facility outside the building.

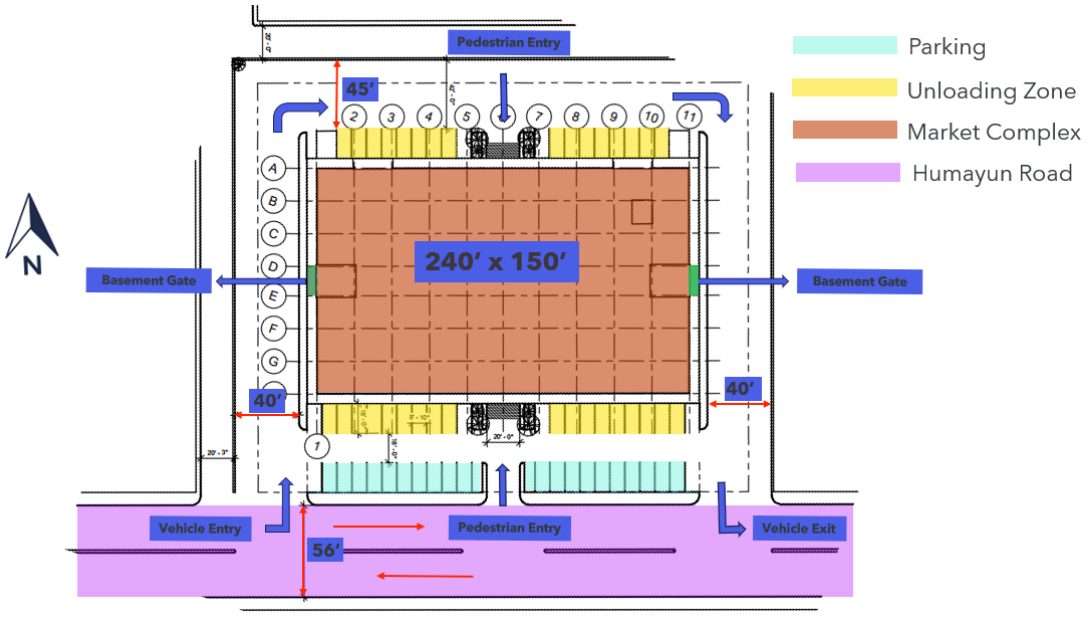


Figure: Masterplan of the project site

The building will be a moment resisting frame system. We will be following BNBC 2020 code. The following important design parameters will be considered,

For seismic design,

Seismic Zone: 2 (Z = 0.2)

Exposer Category: A

Building Type: F

Occupancy Category: III

Additional parameters for wind load,

Exposure: A

Importance factor: 1.15Wind Speed: 65.7 m/s

For fire hazard,

Fire Index: 2

Construction Materials

- Both fine and coarse aggregates will be taken complying with ASTM C33 standard
- Concrete will have strength of 4000 psi and additional steel reinforcement which will be needed will have 60000 psi following ASTM A36 standard.

Preliminary Analysis and Design

We are designing the floor slab with 6″ thickness.

Beam dimension (floor): **12” x 24”**

Beam dimension (stair): **12” x 18”**

Column dimensions (corner): **18” x 18”**

Column dimensions (interior): **30” x 30”**

Column dimensions (edge): **24” x 24”**

Foundation

For foundation we have decided to use raft foundation and 540 shore piles will be needed for the foundation.

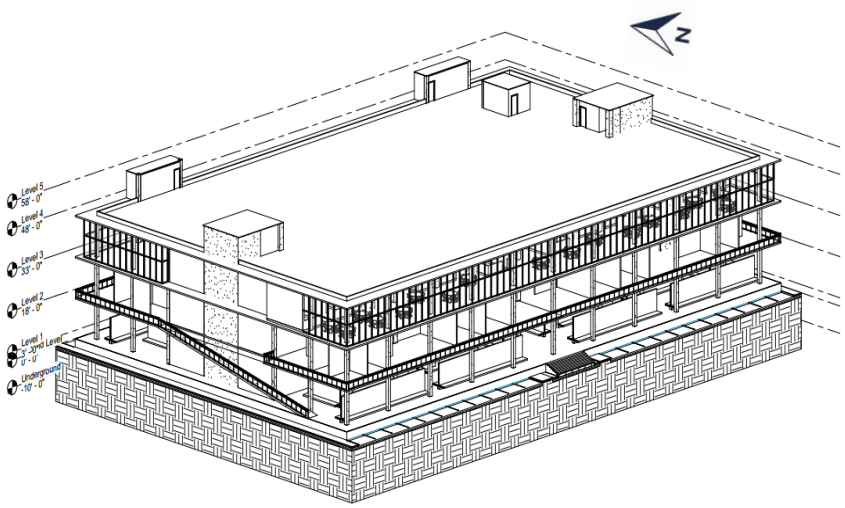


Figure: 3d view of the market

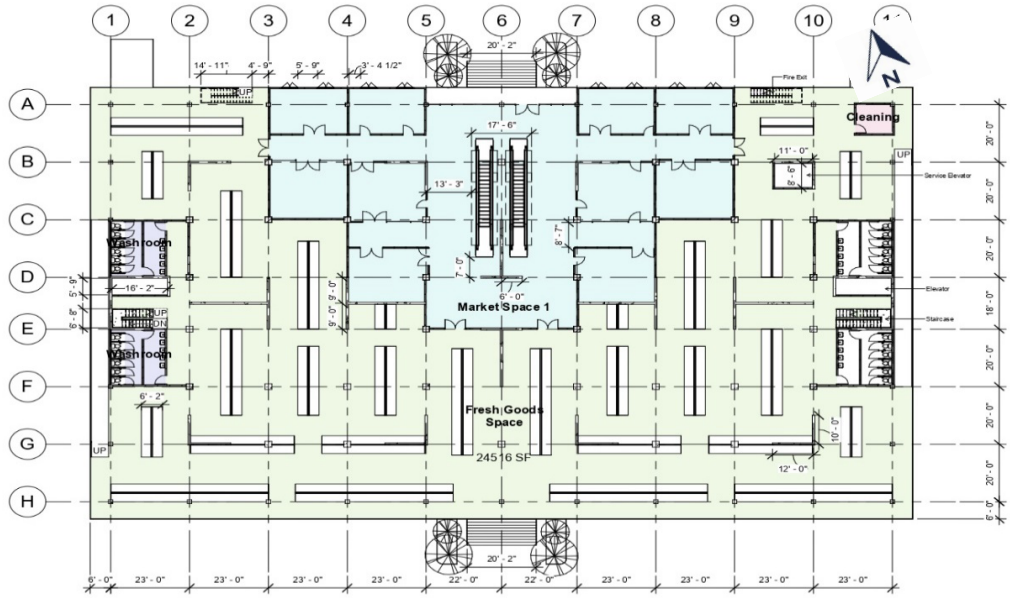


Figure: Plan for Level 1

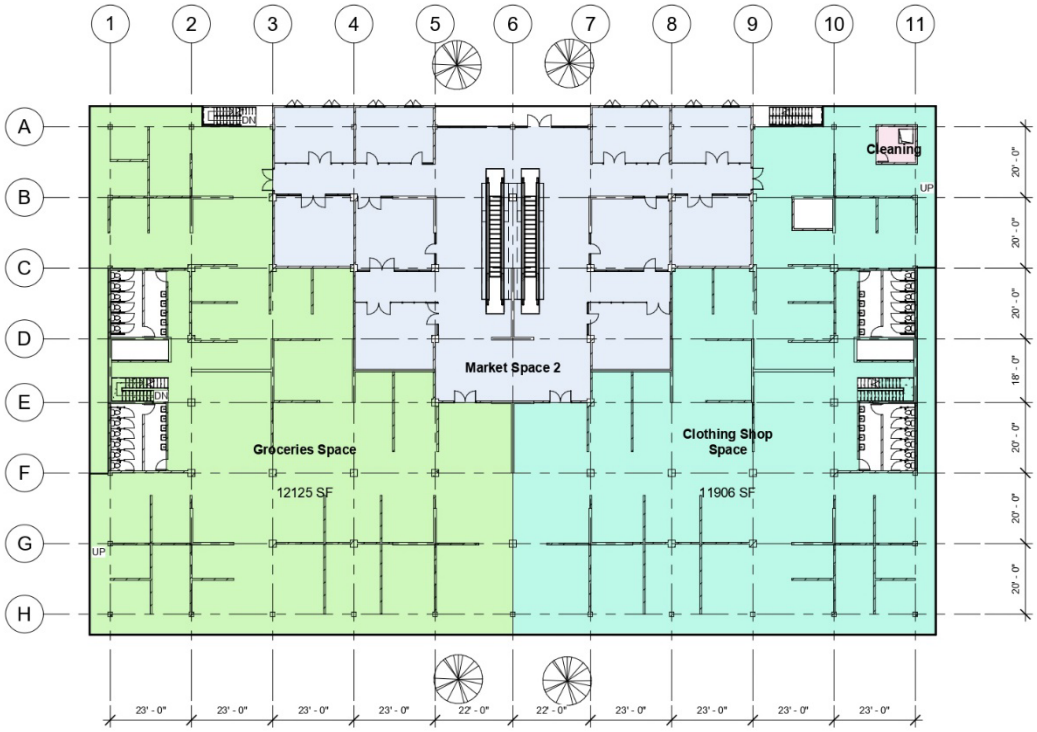


Figure: Plan for Level 2

4.3 Output Plan:

- A kacha bazar(fruit, vegetable market) covering 36,000 sft area.
- Shopping space for clothing and groceries in the first floor with 11,900 sft and 12,000 sft respectively and additional 12,000 sft for markets.
- On the second floor there will be prayer space with 4100 sft and rest of the space will be used for super shop and food courts.
- Three basement that will be used for parking.
- This project will enable employment to people around the market
- It will also be a great hub of economic activities for the area

The Project is supposed to be completed within 2 years. For unforeseen reasons it could be extended by 4 to 6 months. Since this project is relatively conventional, the level of uncertainty is low or moderate. As a resource and time utilization rates will be moderate to high.

4.4 Costs Estimation:

Estimated cost for various parts of the project is given below in BDT.

1.Soil investigation 194,105.00

2.Foundation 127,457,520.00

3. Basement Cost 197,903,969.48

3.Super Structure cost 844,579,076 (Assuming all 8 floors)

4.Additional Super Structure Cost 42,257,915.85

5. Other Building Cost 46,549,502.

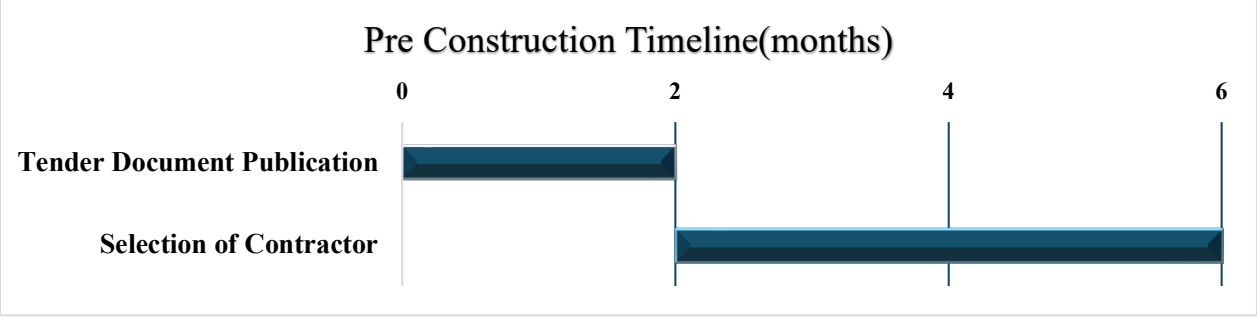
6. Quality, Maintenance Cost 12,589,420.89

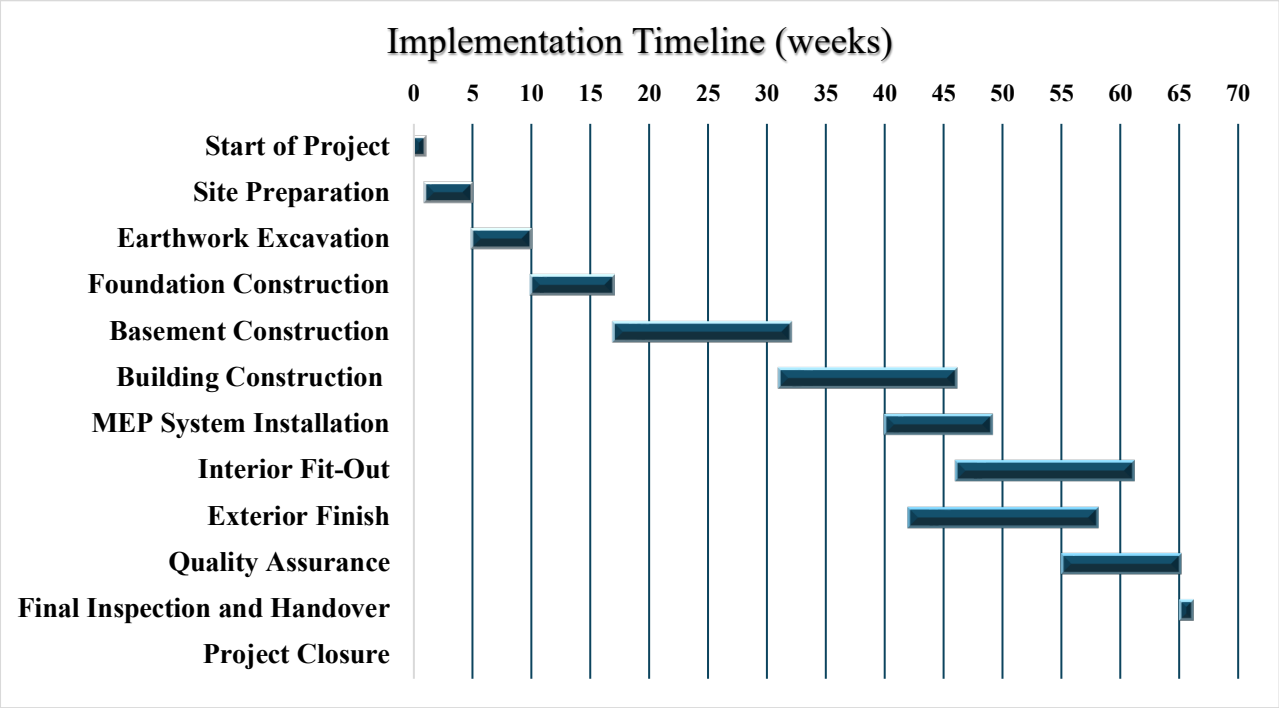
7. Contingency 62,947,104.45

8. Environment Management Plan 55,14,000/-

Grand Total = 138 crores BDT

4.5 Implementation Timeline:





The implementation phase will need around 18 months to complete and after that there will be around 6 more months of monitoring in the post construction phase.

Section 5: Environmental Sustainability, Climate Resilience and Disaster Risk Analysis

5.1 Environmental, Climate Change and Disaster Risk Analysis

5.1.1 Introduction

The Mohammadpur Krishi Market redevelopment project aims to revitalize the fire-damaged market into a modern, multi-storied shopping complex, boosting the local economy. Classified under the red category as per Environmental Protection Rule 2023, the project includes a detailed Environmental Impact Assessment (EIA) involving field visits, identification of parameters, and baseline data analysis. Impacts of project activities during both construction and operational phases on the existing physical, biological, and socio-economic environment were assessed. Mitigation measures will be implemented through an environmental management plan to address identified impacts and ensure sustainable development.

5.1.2 Baseline Environment: Physicochemical

Physical Environment Surrounding Project Site

The physicochemical segment of the baseline environment study typically encompasses specific criteria related to temperature, humidity, seismicity and other critical parameters that helps to visualize the initial environment condition and how it will work out in the project construction. Based on these criteria, we can understand what kind of impact construction will have on the environment and helps in taking measures to reduce those problems.



Fig.: Project Site

Climate

Parameter	Values
Precipitation	1854 mm (yearly)
Relative Humidity	70%
Temperature	28˚C (annual average)
Wind Speed & Direction	66 m/s (from South and South-East)

Seismicity

The site class is SC. Bangladesh is close to one of the most tectonically active regions in the world. It is situated where three tectonic plates namely the Indian plate, the Eurasian plate and the Burmese plate met. Bangladesh has been divided into four seismic zone. Our project location is in Dhaka which is in Zone-2 according to BNBC 2020.

Soil

Due to absence of actual data, a Geotechnical Investigation Report of a Power Grid Substation in Mohammadpur, which is about 1.5 km North-East from our project location, is being used. From the bore-log report, the soil of the site is silty fine sand.

Air Quality

Air quality of the site can be obtained by using air quality monitor and GPS for location of measurement. The current PM2.5 concentration in Dhaka is 5.3 times above the recommended limit given by the WHO 24 hrs. air quality guidelines value.

Noise Level

Noise level of the project site can be obtained by using noise level meter and GPS for location of measurement.

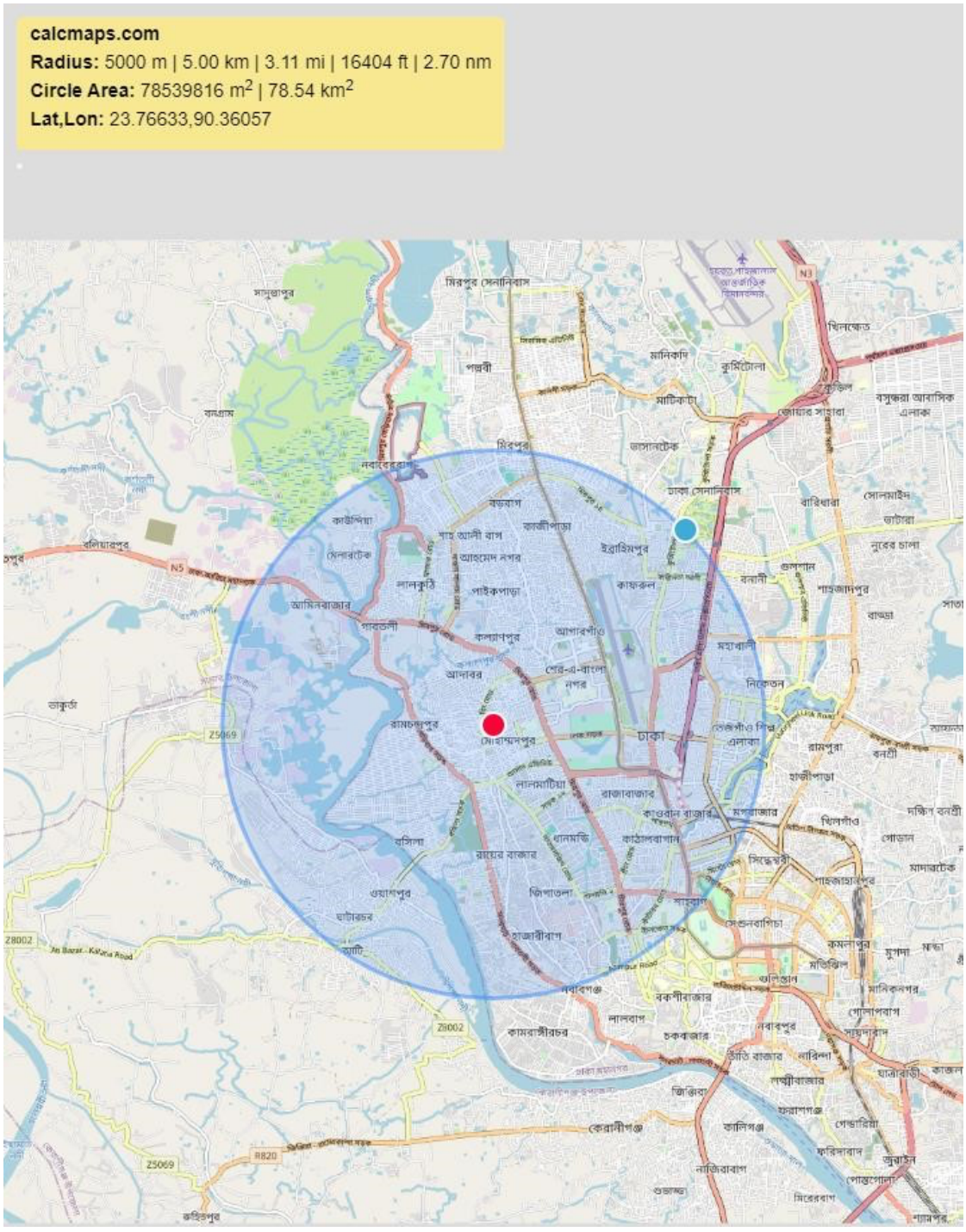
5.1.3 Baseline Environment: Ecological

The Mohammadpur project area, a busy commercial zone, lacks green spaces and biodiversity, with high levels of dust and minimal flora deterring wildlife such as birds. Within a 5 km radius, Dhanmondi Lake suffers from severe pollution due to years of sewage dumping, causing algal blooms that reduce oxygen levels and threaten aquatic life. The lake's poor water quality and lack of cleanliness contribute to the spread of diseases. These environmental conditions highlight the need for effective mitigation measures to address the ecological challenges in the project area.

Methodology

The project’s study area spans a 5-kilometer radius around the site. Ecological impact assessments include bird surveys for biodiversity changes, herpetofauna surveys for reptile and amphibian populations, flora surveys for plant species evaluation, and fish surveys for aquatic biodiversity. Additionally, mammal populations and butterfly diversity are studied to

understand biodiversity changes and habitat health. These systematic observations and documentations inform conservation and management strategies.



Ecological Perspective of the Proposed Project Site: Mohammadpur Krishi Market

The development of a multi-storied shopping complex at the Mohammadpur Krishi Market site requires a strategic ecological approach. This approach should focus on preserving existing green spaces, minimizing habitat disruption, managing waste responsibly, and incorporating eco-friendly infrastructure to balance the ecological impacts. The goal is to ensure that the

project aligns with sustainable development principles, promoting harmony between the new construction and the existing ecosystem.

Baseline Ecological Environment of the Proposed Project and Adjacent Areas

Faunal Diversity:

The Mohammadpur Krishi Market area, situated in an urbanized section of Dhaka, is home to a variety of fauna, including common urban species such as pigeons, crows, sparrows, and various insects. Occasional sightings of small mammals like rats and bats are also noted. Given the urban setting, the faunal diversity is limited but plays a critical role in maintaining urban ecological balance. Comprehensive surveys are essential to document and understand the local faunal diversity, particularly to ensure that the construction does not further disrupt these populations.

Floral Diversity:

Floral diversity in the vicinity of Mohammadpur Krishi Market is primarily characterized by urban vegetation, including street trees, ornamental plants, and patches of greenery in nearby parks and gardens. Common species include neem, banyan, and various flowering plants that contribute to the area's aesthetics and provide essential ecosystem services, such as air purification and shade. Preserving these green spaces and incorporating green design elements into the new complex will be vital in maintaining urban biodiversity.

Threatened Flora and Fauna:

Urbanization in Mohammadpur, including the development of the new shopping complex, poses risks to the already limited natural habitats in the area. While no specific threatened species are exclusively documented at the market site, broader urban development pressures threaten species like the Dhaka Lily and local varieties of bamboo. Habitat degradation, pollution, and lack of green spaces further exacerbate the vulnerabilities of these species. Mitigation strategies, such as creating green rooftops, vertical gardens, and incorporating native plants into landscaping, are recommended to reduce ecological impacts.

Protected Areas, National Parks, Game Reserves, Wildlife Sanctuaries, ECA:

- **Protected Area (PA):** There are no designated protected areas within Mohammadpur, Dhaka.
- **National Park (NP):** Ramna Park, though not located within Mohammadpur, is the closest national park that provides significant green space for the broader Dhaka area.
- **Game Reserve (GR):** There are no game reserves within Mohammadpur or the greater Dhaka metropolitan area.
- **Wildlife Sanctuary (WS):** No wildlife sanctuaries are located within Mohammadpur, Dhaka.
- **Ecologically Critical Area (ECA):** The Buriganga River, designated as an Ecologically Critical Area (ECA), is the nearest ecological zone of significance. The proximity of this river underscores the need for stringent waste management practices to prevent any adverse impacts on this critical water body.

Important Biodiversity Areas and Conservation Implications

Despite the urbanized context of Mohammadpur Krishi Market, the area harbors pockets of biodiversity, particularly in the form of small green spaces, roadside vegetation, and urban gardens. These green areas not only support local fauna but also contribute to the quality of life for residents and visitors. Conservation efforts, even in such an urban setting, are crucial. Strategies should include:

- **Creation of Urban Green Spaces:** Incorporating green spaces within the new market design, such as small parks, green roofs, and vegetated walkways, to support urban wildlife and improve environmental quality.
- **Minimizing Pollution:** Implementing eco-friendly construction practices, waste management systems, and pollution controls to reduce the ecological footprint of the project.
- **Biodiversity Awareness:** Raising awareness among stakeholders, including vendors and customers, about the importance of local biodiversity and sustainable practices.
- **Stakeholder Collaboration:** Engaging with local communities, environmental organizations, and government bodies to foster collaboration in protecting and enhancing urban biodiversity.

The ecological perspective of the Mohammadpur Krishi Market project emphasizes the importance of sustainable development that respects and enhances the existing ecological fabric. By integrating conservation strategies and eco-friendly infrastructure, the project can mitigate potential environmental impacts and contribute positively to the urban ecosystem of Mohammadpur.

5.1.4 Baseline Environment: Socio-economic

5.1.4.1 Introduction

The Mohammadpur Krishi Market, located in a bustling commercial area of Dhaka, serves a diverse population, including local residents, small business owners, and daily commuters. This vibrant marketplace reflects the socio-economic dynamics of urban life in Dhaka, marked by a blend of formal and informal economic activities. Challenges in infrastructure, limited green spaces, and environmental concerns, such as pollution and inadequate waste management, underscore the complexities of this urban setting. However, the market remains a critical hub for trade and employment, highlighting the community's resilience and adaptive capacity amid ongoing socio-economic challenges.

Approach and Methodology

- **Baseline Assessment:** Assess existing socio-economic conditions, including population demographics, economic activities, and access to basic services.
- **Stakeholder Consultation:** Engage with market vendors, local businesses, residents, and government authorities to gather insights and concerns.
- **Economic Impact Analysis:** Evaluate the market's contribution to the local economy, including employment opportunities and income generation.
- **Social Impact Assessment:** Examine the social dynamics of the market, including community interactions, safety, and social cohesion.
- **Infrastructure and Service Provision:** Review the availability and quality of infrastructure and services such as transportation, water supply, and sanitation.
- **Risk Assessment:** Identify potential socio-economic risks, including displacement of informal vendors and increased traffic congestion.
- **Scenario Planning:** Develop potential scenarios to address socio-economic challenges and enhance market operations.

- **Continuous Monitoring and Evaluation:** Establish ongoing monitoring of socio-economic indicators to track progress and impact.

Socio-economic Perspective of Proposed Project and Adjacent Sites

The development and modernization of the Mohammadpur Krishi Market are expected to positively impact the local socio-economic landscape. Upgrading market facilities can enhance business operations, improve vendor conditions, and attract more customers, thereby boosting local commerce. Additionally, improved market infrastructure will provide better access to goods and services, potentially leading to an increase in local employment opportunities and economic empowerment. The project aims to enhance the quality of life for residents and vendors, fostering a more sustainable and vibrant commercial environment.

Findings from Questionnaire Survey

Age Structure: The market area serves a diverse demographic, including middle-aged shop owners, young adults, and elderly residents who rely on the market for daily necessities.

Number of Family Members: The average household size in the Mohammadpur area aligns with the broader trends observed in Dhaka, typically comprising 4 to 5 members per family.

Sex Ratio: The market serves a predominantly male vendor population, but there is a significant presence of female customers and some female-led businesses.

Education: Educational attainment among market vendors varies, with many having basic education levels. The surrounding community benefits from proximity to educational institutions that contribute to local literacy and skill development.

Occupation: The market hosts a variety of occupations, including fruit and vegetable vendors, small shop owners, street food sellers, and informal laborers. These activities are vital for the local economy, providing income for numerous families.

Extent of Respondents' Access to Basic Services

- **Living Standard:** Living standards around the market are mixed, with informal settlements nearby facing challenges such as inadequate housing and sanitation, while other areas have more stable conditions.
- **Electricity Status:** The market and surrounding areas generally have access to electricity, but frequent outages and unreliable supply remain issues for vendors and residents.
- **Water Supply Condition:** Water supply is inconsistent, with some areas experiencing shortages or relying on alternative sources such as bottled or stored water.
- **Sanitation:** Sanitation facilities in the market are limited, with inadequate waste disposal contributing to unhygienic conditions. Efforts are needed to improve sanitation infrastructure and waste management practices.
- **Public Health Status:** Public health is a concern, with reports of communicable diseases linked to poor sanitation and pollution. Access to healthcare services is available, but affordability and quality remain issues.
- **Medical and Health Facility:** The area is serviced by a mix of private clinics and public health facilities, but there are gaps in service quality and availability, particularly for lower-income residents.

Economic Information

- **Wage Rate:** Wage rates for market workers and informal vendors vary, typically ranging from BDT 8,000 to BDT 15,000 per month, depending on the type of work and level of skill. Informal sector workers often face income instability and lack of benefits.

Respondent's Perception Regarding the Proposed Project

Supporters of the market improvement project highlight its potential to enhance local commerce, create jobs, and provide better facilities for both vendors and customers. They see the project as a means to boost the local economy and improve the overall shopping experience.

Opponents express concerns about possible gentrification, displacement of informal vendors, and increased costs that could negatively impact small businesses. There are also worries about potential traffic congestion and disruptions during construction.

Observation of the Survey

Survey findings indicate broad support for the market modernization, with 88% of vendors and residents in favor of the improvements, citing benefits such as increased customer footfall and better working conditions. However, concerns about the affordability of upgraded stalls and the potential exclusion of existing vendors were raised by 45% of respondents. The survey also highlighted that 60% of the surrounding population earns below BDT 1 lakh annually, and 75% rely on informal sector employment, underscoring the need for inclusive and affordable market solutions.

The modernization of Mohammadpur Krishi Market offers a significant opportunity to address socio-economic challenges, but careful planning and community engagement are essential to ensure that the benefits are equitably distributed and that the market remains accessible to all.

5.1.5 Public Consultation and Stakeholder Engagement

5.1.5.1 Introduction

Public consultation and stakeholder engagement are crucial for development of shopping complex, involving diverse stakeholders in discussions, surveys, and forums to gather feedback and address concerns. This transparent and collaborative approach identifies challenges, and generates solutions that satisfies both the demand of stakeholders and customers.

5.1.5.2 Approach and Methodology for Consultation

The consultation approach for shopping complex development involves systematic engagement with stakeholders, using methods like surveys to understand the demands of customers. It is an iterative process, which includes active outreach and feedback integration.

5.1.5.3 Stakeholder Assessment

• Key Informant Interviews (KII) of the Project

.Dhaka North City Corporation (DNCC)

• Mohammadpur Kacha Bazar Traders Association

• Existing shopkeepers

• Local Community

• Buyers

• New shopkeepers

• Retailing Companies

5.1.5.4 Focus Group Discussion (FGD)

• Identifications of Stakeholders

. Conducting surveys

• Feedback Integration

• Reevaluation of decisions

5.1.6 Identification and Assessment of Potential Impacts

5.1.6.1 Introduction

This section identifies and evaluates the potential effects on the physio-chemical, ecological, and socio-economic aspects of the environment within the context of the Mohammadpur Krishi Market redevelopment project, focusing on both the construction and operational phases. Potential impacts have been identified based on baseline data gathered from both secondary and primary sources, considering the unique characteristics of the market environment and the surrounding community.

5.1.6.2 Methodology for Potential Impact Prediction

The methodology for predicting the environmental impacts of the Mohammadpur Krishi Market redevelopment involves scientific analysis, modeling techniques, and stakeholder engagement to support informed decision-making and sustainable development practices. The following qualitative and semi-quantitative prediction tools may be utilized:

- **Expert Opinion:** Leveraging insights from environmental experts, urban planners, and market specialists to assess potential impacts and develop mitigation strategies.
- **Consultations and Questionnaires:** Engaging stakeholders, including market vendors, local residents, and customers, through consultations and surveys to gather input on perceived impacts and community concerns.
- **Checklists:** Utilizing structured checklists to systematically identify potential environmental, social, and economic impacts associated with the project's construction and operational phases.

- **Spatial Analysis:** Applying geographic information systems (GIS) and spatial analysis to assess land use changes, infrastructure impacts, and ecological effects within the market area and its surroundings.
- **Network and Systems Analysis:** Analyzing interconnected systems such as traffic flows, waste management, and utilities to predict and manage potential impacts on urban infrastructure and services.
- **Risk Matrices:** Using risk matrices to prioritize potential impacts based on their likelihood and severity, helping to focus mitigation efforts on the most significant risks.

This methodology provides a comprehensive framework for predicting the potential impacts of the Mohammadpur Krishi Market redevelopment, ensuring that all relevant environmental, social, and economic factors are considered throughout the project's lifecycle.

5.1.6.3 Key Environmental Impacts

Impact During Pre-construction Phase

Environmental Parameters	Magnitudes of Impacts			
	None	Insignificant	Medium	Significant
Land Value Depreciation				
Land Loss				
Traffic Congestion				
Noise Pollution				
Air Pollution				

Impact During Construction Phase :

Physio-chemical Environment

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Air Quality	Air pollution from construction machinery and equipment emissions				

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Water Quality	Water contamination from runoff carrying sediment, chemicals, and construction debris				
Soil Quality	Soil erosion due to land disturbance and excavation activities				

Noise Levels	Noise pollution generated by construction activities such as drilling, blasting, and heavy machinery operation				
Groundwater Quality	Improper storage and disposal of construction materials may pose a risk of groundwater contamination				

Ecological:Environment

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Habitat Disruption	Clearing land for construction can disrupt local wildlife and plant habitats				
Environment al Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Biodiversity Loss	Construction may fragment ecosystems, leading to biodiversity loss				

Wetland Impact	Construction near wetlands can disrupt these sensitive ecosystems				

Socio-economic Environment

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Employment Opportunities	Creation of employment opportunities for local residents, including skilled and unskilled labor positions				
Community Livelihoods	Construction activities may disrupt local businesses affecting community livelihoods				
Utilities	Damaged by excavation or machinery operation during construction				

Impact During Operational Phase

Physio-chemical Environment

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant

Air Quality	Operation of heating, ventilation, and air conditioning systems and increased traffic to and from the market can contribute to air pollution				
Water Quality	Disposal of wastewater from market facilities can impact water quality				
Soil Quality	Intensive land use for construction and facilities may continue to impact soil quality				
Noise Levels	Ongoing campus activities and events can contribute to noise pollution				
Groundwater Quality	Excessive groundwater extraction for campus operations may impact groundwater levels				
Waste Generation	Campus activities can lead to the generation of solid waste				

Ecological Environment

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Wildlife	Disruption of local wildlife populations due to increased human activity and introduction of invasive species through importation of plants				
Soil Erosion	Soil erosion and degradation from inadequate landscaping practices or heavy foot traffic in outdoor areas				
Biodiversity Loss	Loss of biodiversity and ecosystem function due to changes in land use patterns and urbanization associated with market operations				

Socio-economic Environment

Environmental Parameters	Impacts	Magnitudes of Impacts			
		None	Insignificant	Medium	Significant
Employment Opportunities	Market operations may provide ongoing employment, positively impacting the local job market.				
Economic Growth	Market activities contribute to the local economy				
Social Services	The market population may contribute to increased demand for social services, impacting local service providers				
Infrastructure Demand	Ongoing operations may lead to increased demand for transportation and utilities				

5.1.7 Mitigation Measures

Introduction

Mitigation of impacts, within the Environmental Impact Assessment (EIA) process, involves identifying potential measures to address significant adverse impacts. Effective mitigation measures can improve the quality of life for surrounding communities by minimizing air and noise pollution, water contamination, and other disruptions related to construction and operation activities.

Mitigation Measures for Impact During Construction Period:

Issue	Proposed Mitigation Measures
Air Pollution	Use low-emission construction machinery and equipment.
	Implement dust control measures, such as water spraying on construction sites and covering materials.
Water Pollution	Store chemicals and construction materials away from water bodies to prevent spills.
	Use spill prevention measures and train workers on spill response.
Soil Pollution	Implement erosion control measures like silt fences and straw mulching to prevent soil erosion.
	Limit heavy machinery in sensitive areas to prevent soil compaction.
Noise Pollution	Schedule noisy construction work during daytime hours and limit during early morning or late evening.
	Use noise-reducing construction equipment when possible.
Waste Disposal	Utilize environmentally friendly materials with low toxicity and minimize hazardous emissions.
	Train workers on proper waste management and disposal methods.
Occupational Health and Safety	Enforce strict safety protocols and provide regular safety training for workers, including PPE usage.

Disruption of Utility Services	Conduct utility surveys before construction to identify underground utilities and provide temporary services as needed.

Mitigation Measures for Impact During Operation Period:

Issue	Proposed Mitigation Measures
Drainage Congestion	Regularly clean and maintain drainage systems to prevent blockages.
	Ensure proper landscaping to reduce sedimentation in drainage channels.
Water Pollution	Establish a monitoring program to assess water quality and adjust pollution prevention plans as needed.
	Promote sustainable landscaping practices, minimizing the use of chemical fertilizers and pesticides.
Waste Generation	Implement waste management programs and encourage recycling among vendors and visitors.
Traffic Congestion	Develop traffic management strategies to reduce congestion, such as optimizing entry and exit points.
Increased Demand for Services	Collaborate with local authorities to ensure adequate utility services and implement energy-efficient solutions.

These mitigation measures are designed to minimize the environmental impacts associated with the construction and operation of the Mohammadpur Krishi Market project, contributing to a more sustainable and resilient infrastructure development.

5.1.8 Environmental Management Plan (EMP)

The Environmental Management Plan (EMP) for the Mohammadpur Krishi Market project establishes a framework for monitoring, reporting, and managing environmental performance throughout both the construction and operation phases. The EMP aims to ensure regulatory compliance, promote sustainable practices, minimize environmental impacts, and enhance the well-being of the market and its surrounding community.

Work Plans and Schedules

As part of the construction contract, the Contractor will develop an implementation schedule for environmental management and monitoring, aligning with recommended mitigation measures for identified significant impacts. This will include specific timelines and responsibilities for executing environmental protection measures during the construction phase.

Monitoring Program

Pre-Construction Phase

- **Baseline Assessment:** Conduct baseline surveys to document existing environmental conditions.
- **Regulatory Compliance Check:** Ensure all project activities comply with relevant environmental regulations and standards.
- **Community Engagement:** Engage with local stakeholders to inform and address concerns related to the project.
- **Monitoring Protocol Development:** Develop protocols for monitoring key environmental parameters.
- **Data Management and Reporting:** Establish systems for data collection, analysis, and reporting to track environmental performance.

Construction Phase The primary focus during the construction phase will be on monitoring the following environmental aspects:

- **Air Quality:** Regular monitoring of particulate matter and emissions from construction activities.
- **Noise Level:** Measure noise levels to ensure they remain within acceptable limits, particularly near residential areas.
- **Ground Water Quality:** Monitor potential contamination from construction activities, especially near water sources.
- **Surface Water Quality:** Monitor water quality in nearby bodies, such as Dhanmondi Lake, to prevent pollution from runoff.
- **Soil Quality:** Regular checks to prevent soil erosion and contamination.

- **General Site Condition:** Ensure that the site remains clean and organized, minimizing environmental disturbance.
- **Construction Safety Measures:** Implement safety protocols to protect workers and surrounding communities.
- **Stakeholder Engagement:** Maintain ongoing communication with stakeholders to manage expectations and address concerns.

Operation Phase During the operation phase of Mohammadpur Krishi Market, environmental monitoring will focus on:

- **Noise Level:** Monitor noise pollution from market operations and take measures to reduce impacts on the surrounding community.
- **Ambient Air Quality:** Regular checks to maintain air quality standards within and around the market.
- **Soil Quality:** Monitor potential contamination from market activities, including waste disposal and spillages.
- **Solid Waste Management:** Implement and monitor effective waste management practices, including recycling and safe disposal.
- **Traffic Congestion Management:** Develop strategies to manage vehicle flow and reduce congestion in the market area.
- **Operation Safety Measures:** Ensure that safety protocols are in place and regularly updated to protect vendors and visitors.
- **Stakeholder Engagement:** Continue engagement with the community and stakeholders to foster support and address any ongoing environmental concerns.

The EMP is designed to ensure that the environmental impacts of the Mohammadpur Krishi Market project are effectively managed, contributing to the sustainable development of the market and the surrounding area. Regular reviews and updates of the EMP will be conducted to incorporate lessons learned and respond to changes in environmental conditions or regulations.

Preliminary cost estimates for monitoring and mitigation activities during construction phase:

Parameter/Activity	**Frequency of activity**	**Cost in BDT (per month)**	**Cost in BDT (18 Months)**
Particulate Matter (PM10, PM2.5)	Once every month	40,000/- per each set of PM10 and PM2.5 measurement	7,20,000/-
Noise Level (ambient and personal exposure)	Once every month (day and night)	40,000/- (per set of measurement)	7,20,000/-
Surface water quality (parameters: Turbidity, Total Suspended Solids, BOD5, Dissolved Oxygen)	1measurement be for estarting construction, then in each subsequent month	Tk. 10,000/- (Per set of measurement)	1,90,000/-
Drinking water quality testing: pH, color, turbidity, total hardness, chloride, Total and Fecal coliform, Total Dissolved Solids, Arsenic, Iron, Manganese, Electrical Conductivity (salinity), Free Chlorine	Monthly	Tk. 12,000/- per set of measurement	2,16,000/-
Soil Quality (Heavy metals Pb, Cr, Cd)	Monthly	Tk. 6,000/- per set of measurement	1,08,000/-
Site Cleaning and preparation including providing necessary protective fencing and safety measures with sign boards.	Periodic	Lump sum	5,00,000/-

Vegetation & Tree plantation around the site including fencing/ conservation/ maintenance for 2 years. Trees need to be replanted around the periphery of the proposed site at an interval of 10 feet @ Tk. 1500 for each tree.	Periodic	Lump sum	2,00,000/-
Providing safety gear package like hand gloves, eye protection glasses, helmets, rubber shoes, light reflecting dress etc. for 40 ets @ Tk. 10,000 for each set	-	Lump sum	4,00,000/-
Drinking water container for workers including necessary ceramic filters for providing drinkable water	-	Lump sum	1,00,000/-
Temporary Sanitary Latrine/ Septic Tank/ Portable Toilet: 2 nos. (1 no of Toilet for female and 1 no of Toilet for male) @Tk. 50,000	-	Lump sum	1,00,000/-
Waste disposal charge from site by outsourced cleaners	Daily or weekly collection of solid waste	20,000/-	3,60,000/-
Dust suppression measures like water sprinkling on aggregates/ unpaved roads, in and around the work site (Lump Sum). For road construction works cost of this item has been mentioned in road section.	Daily or weekly activities	Lump sum	4,00,000/-
Health and safety warning sign	-	Lump sum	20,000/-
Appointment of an Environmental Health	-	50,000/-	9,00,000/-

and Safety officer for Environmental and Social Management and Monitoring during construction (salaried position)			
Total			49,34,000/-

Preliminary cost estimates for monitoring and mitigation activities during operational phase:

Parameter/Activity	**Frequency of activity**	**Cost estimate (per month)**	**Cost in BDT (6 Months)**
CO, SO_X, NO_X, PM_{10}, SPM for generator stack emission	Monthly	Tk. 40,000/-	2,40,000/-
Treated water for drinking: pH, color, turbidity, total hardness, chloride, Total and Fecal coliform, Total Dissolved Solids, Arsenic, Iron, Manganese, Electrical Conductivity (salinity), Free Chlorine	Monthly	Tk. 20,000/-	1,20,000/-
Noise level monitoring at surrounding areas (nearby residential and market areas). Noise emissions from generator.	As and when required	Administration conducts monitoring and records positions by Cable Car Project- owned noise level meter and GPS (cost of standard noise level meter 50,000/- and GPS is 10,000/-)	1,00,000/-
Solid waste management, recycling of wastes	Monthly	Tk. 10,000/-	60,000/-

Fire drills	Monthly	Contracted out to Fire Department or other competent contractors	60,000/-
Total			5,80,000/-

Total Cost of Monitoring	55,14,000/-

5.2 Assessment of Disaster Resilience of the Project

Contingency Plan for Emergency Disaster Management:

A comprehensive emergency management plan includes designated evacuation routes and assembly points in case of fire, earthquake, or flood. An institutional arrangement will be established to manage the shutdown of utilities like gas, electricity, and water during emergencies. The plan also includes training for personnel on emergency response protocols.

5.2.1 Business Continuity Plan:

Key response and recovery priorities involve the swift restoration of critical services, such as power, water, and access to the building. A backup power supply via generators and water storage systems will support the continuity of operations.

5.2.2 Time of Recovery:

The projected recovery time after a disaster event is approximately 3-4 weeks, depending on the severity of the damage. This timeline includes structural assessments, utility restoration, and resuming normal operations.

5.2.3 Reporting of Residual Risks:

Despite mitigation measures, residual risks include potential minor flooding during extreme weather events and limited seismic damage. Continuous monitoring and annual reviews of the risk management plans will be conducted to address any evolving threats.

5.3 Traffic Impact Assessment

5.3.1 Introduction

Background

Bangladesh faces significant transportation-related challenges, especially in heavily populated cities like Dhaka. Bangladesh, a developing nation aiming for rapid growth and urban development, has transportation issues. Because of inadequate planning by authorities and noncompliance with development regulations, these problems frequently get worse when new infrastructure is developed.

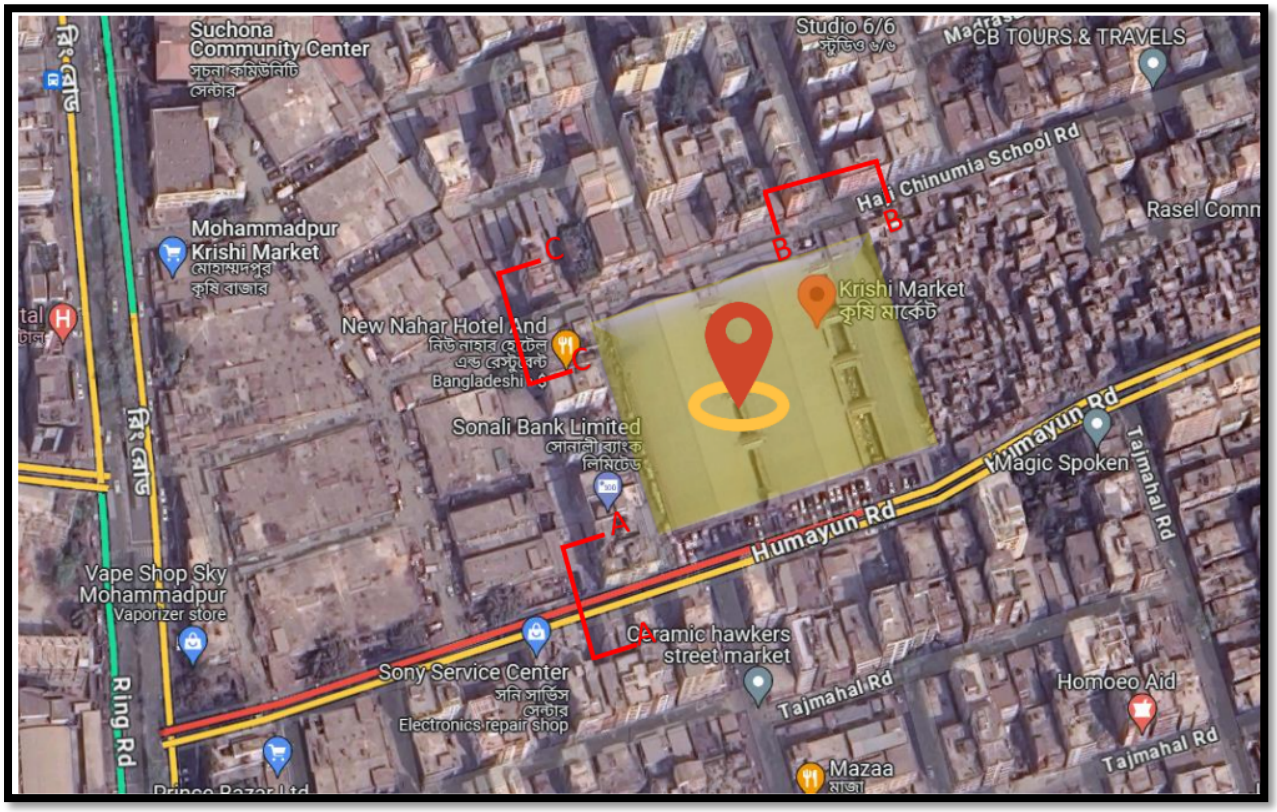
An essential procedure for assessing the state of transportation in a planned development area and projecting future traffic scenarios after construction is the Traffic Impact Assessment (TIA). To ensure sustainable development, TIA is usually carried out for locations with high trip generation, such as parks, schools, markets, hospitals, and shopping centers.

Building new infrastructure in Dhaka is particularly difficult because of the city's congested road system. The purpose of this report is to evaluate the possible traffic impact of a proposed multi-story shopping complex at Mohammadpur Krishi Market and suggest mitigation strategies for any negative effects.

Existing condition:

The Site:

Mohammadpur Krishi Market a market of agricultural commodities established on an area of 5 acres at Mohammadpur, Dhaka in 1978 under the ownership of the Directorate of Agricultural Marketing of Bangladesh. The market is bounded by the Shiah Mosque on the south and the Humayun road on the west.



5.3.2 Traffic Volume with PCU:

A-A Section:

Traffic Survey			
Section	Types of Vehicles	Peak Hour Volume	Peak Hour Volume (Converted to PCU)
A-A	Rickshaw	550	440
	Motorbike	200	60
	Bicycle	70	14
	CNG	40	20
	Car	80	80
	Van	276	221
	Pickup Truck & Truck	45	135
		Total	970

Roadway Capacity Calculation (Highway Capacity Manual)

Given,

For **Humayun Road**,

2 lane, 2 way

Operating speed 30 mph, LOS E

Lane width 10 ft

Roadway Capacity=Capacity from Table 1 * factor for LOS (from table 2) * factor for lane width (from table 3) * factor for shoulder width (from table 4)

Table - 1

Roadway type	Capacity (passenger veh/h)
Multi-lane Free flow	2000 per lane
Two lane, two way	2000 total both direction
Three lane, two way	4000 total both direction

Table-2**

Level of Service	Traffic flow condition: Description	Traffic flow condition: Operating speed (mph)	Passing sight distance > 1500 ft	Basic limiting value for AHS* of 70 mph	Maximum service volume under ideal condition
A	Free flow	= > 60	100	0.20	400
			80	0.18	
			60	0.15	
			40	0.12	
			20	0.08	
			0	0.04	
B	Stable	=>50	100	0.45	900
			80	0.42	
			60	0.38	
			40	0.34	
			20	0.30	
			0	0.24	
C	Approaching unstable flow	=>35	100	0.7	1400
			80	0.68	
			60	0.65	
			40	0.62	
			20	0.59	
			0	0.54	
D	Approaching unstable flow	= > 35	100	0.85	1700
			80	0.84	
			60	0.83	
			40	0.82	
			20	0.81	
			0	0.80	
E	Unstable flow	30	Not applicable	<=<1	2000
F	Forced Flow	<30	Not Applicable	Not Meaningful	Widely variable (0 to capacity)

* AHS = Average highway speed ** Source: Highway Capacity Manual (FHA, USA)

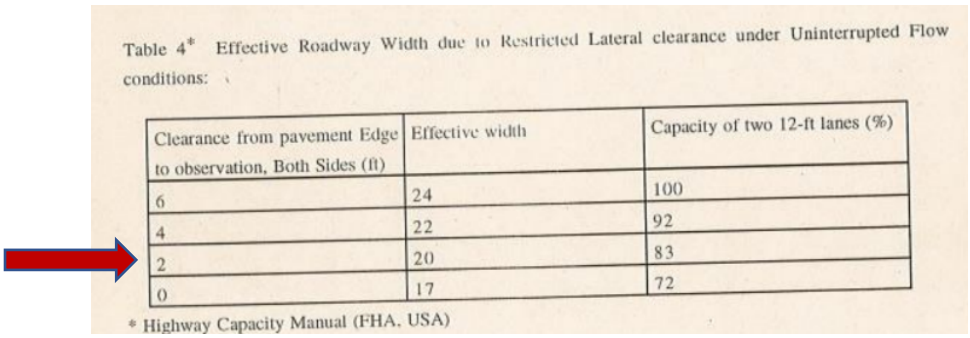


Table 4* Effective Roadway Width due to Restricted Lateral clearance under Uninterrupted Flow conditions:

Clearance from pavement Edge to observation, Both Sides (ft)	Effective width	Capacity of two 12-ft lanes (%)
6	24	100
4	22	92
2	20	83
0	17	72

* Highway Capacity Manual (FHA, USA)

Road Capacity = 1000*0.81*1*0.83

=672 pcu/hr/lane

=672*2 pcu/hr

=1344 pcu/hr (2 lanes)

B-B Section:

Traffic Survey			
Section	Types of Vehicles	Peak Hour Volume	Peak Hour Volume (Converted to PCU)
B-B	Rickshaw	360	288
	Motorbike	120	36
	Bicycle	48	10
	CNG	37	19
	Car	55	55
	Van	159	127
	Pickup Truck	25	75
		Total	609

Roadway Capacity Calculation (Highway Capacity Manual)

Given,

For **B-B Section**,

2 lane, 2 way

Operating speed 30 mph, LOS E

Roadway Capacity=Capacity from Table 1 * factor for LOS (from table 2) * factor for lane width (from table 3) * factor for shoulder width (from table 4)

Table - 1

Roadway type	Capacity (passenger veh/h)
Multi-lane Free flow	2000 per lane
Two lane, two way	2000 total both direction
Three lane, two way	4000 total both direction

Table 3* Effect of lane width on capacity for uninterrupted flow conditions

Lane Width (ft)	Capacity %	
	Tow-lane Roadways	Multi-lane Roadways
12	100	100
11	88	97
10	81	91
0	76	81

* Highway Capacity Manual (FHA. USA)

		(mph)	1500 ft	AHS - of 70 mph	condition
A	Free flow	= > 60	100	0.20	400
			80	0.18	
			60	0.15	
			40	0.12	
			20	0.08	
			0	0.04	
B	Stable	=>50	100	0.45	900
			80	0.42	

Table 4* Effective Roadway Width due to Restricted Lateral clearance under Uninterrupted Flow conditions:

Clearance from pavement Edge to observation, Both Sides (ft)	Effective width	Capacity of two 12-ft lanes (%)
6	24	100
4	22	92
2	20	83
0	17	72

* Highway Capacity Manual (FHA. USA)

D	Approaching unstable flow	= > 35	100	0.85	1700
			80	0.84	
			60	0.83	
			40	0.82	
			20	0.81	
			0	0.80	
E	Unstable flow	30	Not applicable	<=<1	2000
F	Forced Flow	<30	Not Applicable	Not Meaningful	Widely variable (0 to capacity)

* AHS = Average highway speed ** Source: Highway Capacity Manual (FHA, USA)

Road Capacity = 1000*1*0.76*0.72

=547 pcu/hr/lane

=547*2 pcu/hr

=1094 pcu/hr (2 lanes)

C-C Section:

Traffic Survey			
Section	Types of Vehicles	Peak Hour Volume	Peak Hour Volume (Converted to PCU)
C-C	Rickshaw	335	268
	Motorbike	112	34
	Bicycle	39	8
	CNG	31	16
	Car	47	47
	Van	152	122
	Prickup Truck & Truck	28	84
		Total	578

Roadway Capacity Calculation (Highway Capacity Manual)

Given,

For **C-C Section**,

2 lane, 2 way

Operating speed 30 mph, LOS E

Roadway Capacity=Capacity from Table 1 * factor for LOS (from table 2) * factor for lane width (from table 3) * factor for shoulder width (from table 4)

Table - 1

Roadway type	Capacity (passenger veh/h)
Multi-lane Free flow	2000 per lane
Two lane, two way	2000 total both direction
Three lane, two way	4000 total both direction

Table-2**

Level of Service	Traffic flow condition		Passing sight distance > 1500 ft	Basic limiting value for AHS* of 70 mph	Maximum service volume under ideal condition
	Description	Operating speed (mph)			
A	Free flow	= > 60	100	0.20	400
			80	0.18	
			60	0.15	
			40	0.12	
			20	0.08	
			0	0.04	
B	Stable	=>50	100	0.45	900
			80	0.42	
			60	0.38	
			40	0.34	
			20	0.30	
			0	0.24	
C	Approaching unstable flow	=>35	100	0.7	1400
			80	0.68	
			60	0.65	
			40	0.62	
			20	0.59	
			0	0.54	
D	Approaching unstable flow	= > 35	100	0.85	1700
			80	0.84	
			60	0.83	
			40	0.82	
			20	0.81	
			0	0.80	
E	Unstable flow	30	Not applicable	<=<1	2000
F	Forced Flow	<30	Not Applicable	Not Meaningful	Widely variable (0 to capacity)

* AHS = Average highway speed ** Source: Highway Capacity Manual (FHA, USA)

Table 4* Effective Roadway Width due to Restricted Lateral clearance under Uninterrupted Flow conditions:

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0	17	72

* Highway Capacity Manual (FHA, USA)

Road Capacity = 1000*1*0.76*0.72

= 547 pcu/hr/lane

= 547*2 pcu/hr

= 1094 pcu/hr (2 lanes)

2.3 Parking Conditions:

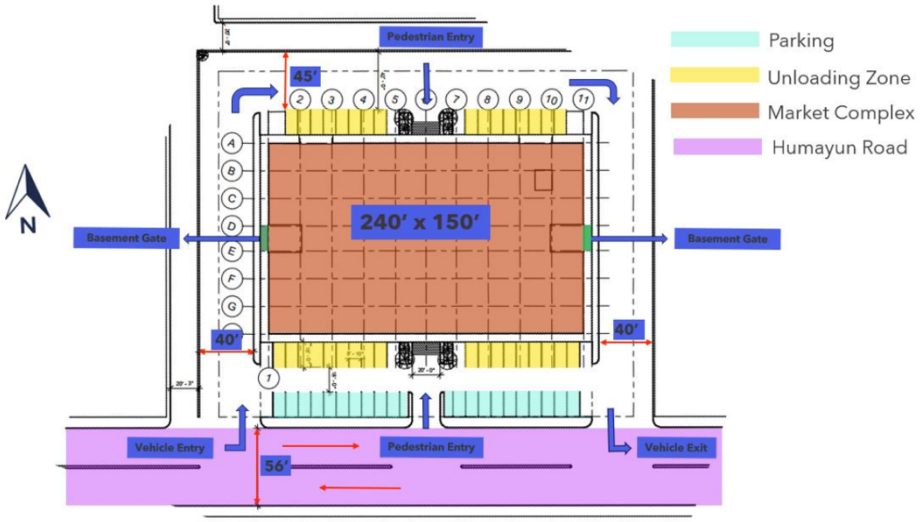
In existing conditions there are no parking facilities.

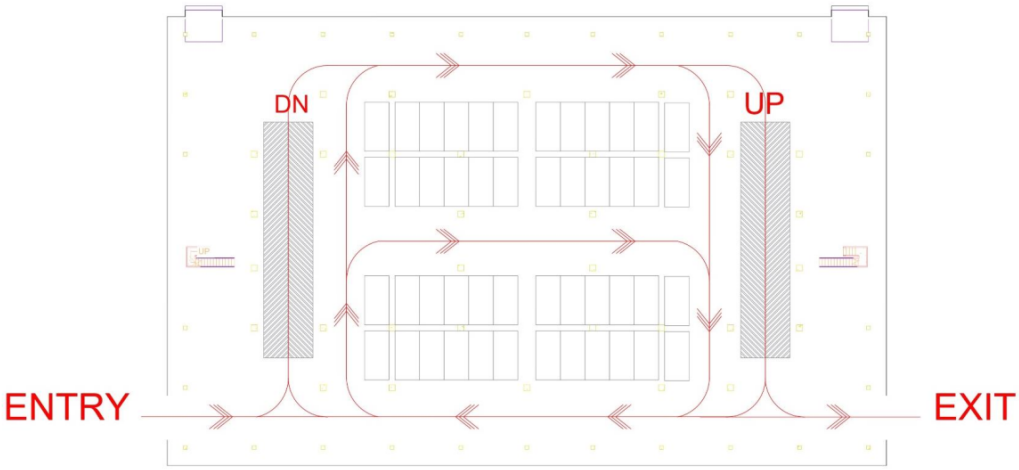
2.4 Footpath Conditions:

In existing conditions there are no footpath facilities.

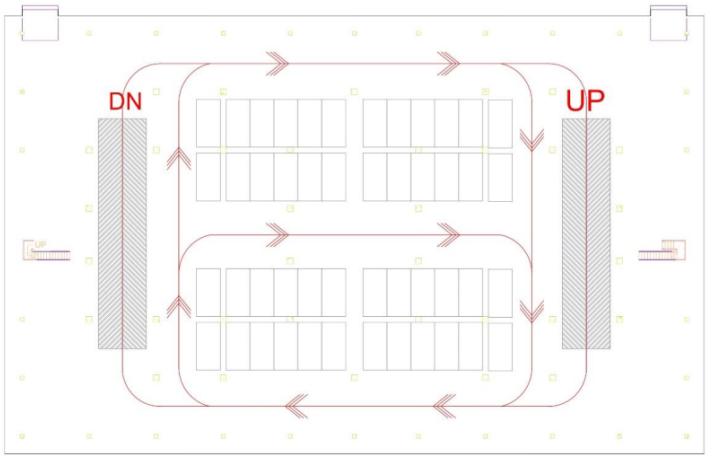
3. Development Plan:

Masterplan:

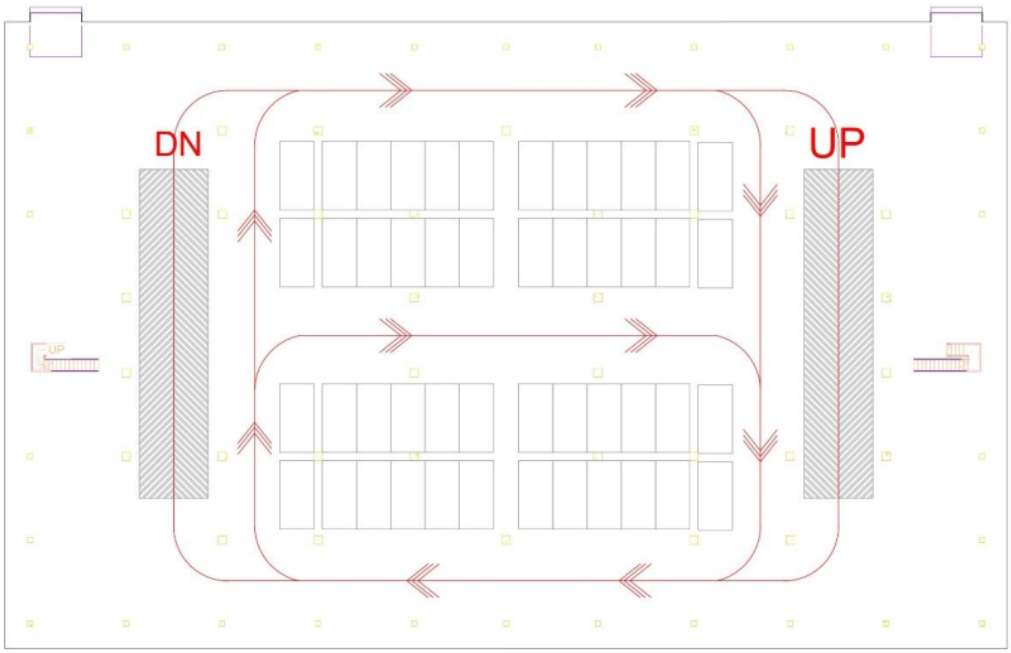




First Basement



Second Basement



Third Basement

1,2 floor shop || 3rd floor=Shop+Resturant || 4,5,6,7,8th floor = Office

FOR SHOPS:

<u>1 car parking for 200m^2(2160ft$^{2)}$ Gross area:</u>

1st floor= 240X150= 36000 ft^2

2nd floor= 240X150= 36000 ft^2

3rd floor= 240X84= 36000 ft^2

TOTAL = 92160ft^2

Car parking need for shops= 92160/2160

= 42.67

=43 no

FOODCOURT/RESTURANTS:

1 car parking per 100m^2(1080ft^2) gross area:

3rd floor=240X66=15840ft^2

Car parking need for food court= 15840/1080

=14.67

=15no

OFFICE:

1 car parking for 200m^2(2160ft$^{2)}$ Gross area:

4th, 5th, 6th, 7th, 8th floor=5X(240X150)=180000ft^2

Car parking need for office= 180000/2160

=83.33

=84no

Total car parking needed= 43+15+84

= 142no

Parking spaces for motorcycles and rickshaws will be available.

Additionally, pedestrian entry will be made in order to improve the walkway scenarios.

Traffic Impact Assessment (TIA):

Trip Generation:

For predicting the trips generated by the proposed market we used trip attraction model established by the study of similar shopping centres in Dhaka city-
Ref: DETERMINATION OF TRIP ATTRACTION RATES OF SHOPPING CENTERS IN DHAKA CITY

December 2014

DOI:10.13140/RG.2.1.2955.7522

Conference: 2nd International Conference on Advances in Civil Engineering 2014 (ICACE-2014) At: CUET, Chittagong, Bangladesh

Primary Model:

$Y = 34.036 + 0.10\ X1 + 53.984\ X2 + 23.778\ X3.$

$R^2 = 0.850$; Adjusted $R^2 = 0.625$

Where X1, X2 and X3 are the total numbers of stores, availability of parking and restaurant respectively.

and Y is the average trip attractions of shopping centre with respect to number of persons per 15 minutes.

For our proposed shopping centre-

Total numbers of stores X1= 294

Availability of parking X2 = 1

Availability of restaurant X3 = 1

Average trip attractions of shopping centre with respect to number of persons per **15 minutes**
Y = 141.198

565 trips per hour

Alternate Model:

$Y = 35.480 + 0.077\ X1 + 63.136\ X2 + 23.739\ X3;$

$R^2 = 0.786$; Adjusted $R^2 = 0.464$

Where X1, X2 and X3 are the gross floor area per 1000 sq ft, availability of parking and restaurant respectively and Y is the average trip attractions of shopping center with respect to number of persons per 15 minutes

For our proposed shopping centre-

Total numbers of stores X1= 294

Availability of parking X2 = 1

Availability of restaurant X3 = 1

Average trip attractions of shopping centre with respect to number of persons per **15 minutes** Y = 141.198

4.2 Trip Distribution:

From our survey of the customers, we found about 55**%** trips are generated near areas adjacent to Haji Chinumia School Road (both section B-B and C-C combined) road for and other 45**%** generated from areas adjacent Humayun Road.

Direction	From/To Humayun Road	From/To Haji Chinumia School Road
% Distribution	45	55
Distributed Trip (Total 1046)	471	575

Mode Distribution:

From our survey we found out that customers of the current market use 4 modes of vehicles. The percentages of the modes for Humayun road are-
Car- 8
CNG- 2
Rickshaw- 45
Motorbike- 6
Van- 23
Pickup Truck-14
Bicycle-1

Areas adjacent to Humayun Road - 970 PCU **per hour**				
Type of Vehicle	**Mode Share**	**No of Vehicles**	**PCU Value**	**PCU/hr**
Car	8	80	80	80
CNG	2	40	20	20
Rickshaw	45	550	440	440
Motorbike	6	200	60	60
Van	23	276	221	221
Pickup Truck	14	45	135	135
Bicycle	1	70	14	14
				Total- 970
Haji Chinumia School Road - 1189 PCU per hour				
Type of Vehicle	**Mode Share**	**No of Vehicles**	**PCU Value**	**PCU/hr**
Car	9	102	102	102
CNG	3	68	35	35
Rickshaw	47	695	556	556
Motorbike	6	222	70	70
Van	21	311	249	249
Pickup Truck	13	53	159	159
Bicycle	2	87	18	18
				Total- 1189

The percentage of the modes for Haji Chinumia School Road are-

Car-9

CNG-3

Rickshaw-47

Motorbike- 6

Van- 21

Pickup Truck-13

Bicycle-2

Future Traffic Projection:

Current estimates suggest that the vehicle growth rate in Dhaka hovers around 5-10% annually (Ref: https://link.springer.com/article/10.1007/s12469-021-00270-4) , depending on the type of vehicle and area We are considering a growth of 5% to calculate future traffic demand

Humayun road

Year	2025	2030	2035	2040	2045
Projected peak hour traffic volume (pcu/hr)	495	631	806	1028	1312
Peak hour traffic generated from the shopping mall (pcu/hr)	1019	1300	1659	2117	2702
Forecasted peak hour traffic volume (pcu/hr)	**1514**	**1931**	**2465**	**3145**	**4014**

Haji Chinumia School Road Section B-B and Section C-C

Year	2025	2030	2035	2040	2045
Projected peak hour traffic volume (pcu/hr)	604	771	983	1255	1602
Peak hour traffic generated from the shopping mall (pcu/hr)	1248	1593	2034	2595	3313
Forecasted peak hour traffic volume (pcu/hr)	**1852**	**2364**	**3017**	**3850**	**4915**

The tables present forecasted peak hour traffic volumes on two roads, Humayun Road and Haji Chinumia School Road Road, considering projected traffic and additional traffic generated by a shopping mall.

Haji Chinumia School Road:

- 2025-2046: Projected peak hour traffic volume increases from 1248 to 3313 pcu/hr.
- Traffic from the shopping mall: Starts at 131 pcu/hr in 2025, increasing to 1602 pcu/hr by 2046.
- Total forecasted peak hour traffic volume: Grows from 1852 pcu/hr in 2025 to 4915 pcu/hr in 2046.

Humayun Road:

- 2025-2046: Projected peak hour traffic volume increases from 1019 to 2702 pcu/hr.

- Traffic from the shopping mall: Starts at 495 pcu/hr in 2025, increasing to 1312 pcu/hr by 2046.
- Total forecasted peak hour traffic volume: Grows from 1514 pcu/hr in 2025 to 4014 pcu/hr in 2046.

Recommendations:

From comparison between roadway capacity and future traffic projection, the proposed shopping complex will impact the surrounding roads from as early as 2025. Because, after this period, the forecasted peak hour traffic volume will exceed the current road capacity. To ensure safety and comfort of users of the shopping complex, the following steps are recommended:

1. Development of footpaths to accommodate customers arriving on foot from nearby locations.
2. Provision of drop zones for customers arriving using different modes of transportation.
3. Strict enforcement of laws to remove obstacles that hamper traffic flow.
4. Implementation of demand responsive traffic signals to improve flow and reduce wait times during peak hours.

Based on future assessment, more steps should be taken to address future needs.

Conclusion:

From this Traffic Impact Assessment (TIA) for the proposed multi-storied shopping complex, following conclusions can be drawn:

1. The construction of the shopping complex is expected to have an impact on the surrounding roads from 2025, as current road conditions and infrastructure capacity will be insufficient to accommodate the projected traffic demand of 2025.
2. Recommendations provided in this TIA such as development of footpaths, provision of drop zones should be adequate for the surrounding roads to handle the new traffic generated by the shopping complex.

3. The proposed number of parking spaces should be adequate to fulfill the increased demand for parking due to the shopping complex.

Dhaka, the capital of Bangladesh, is still under development and already a densely packed city. To ensure proper development, Traffic Impact Assessment is crucial to assess the impact of a project and conclusion of mitigation measures within the plan of development. If the recommended measures are effectively implemented, the construction of the new shopping complex should not negatively affect the traffic conditions of that area.

Section 6: Cost-Benefit Analysis

6.1 Financial Analysis

The financial costs of the associated project include two main costs. These are-

Construction costs

These costs include construction worker costs, equipment operating cost, safety protocol installation cost, material costs.

Maintenance costs

These costs include Security cost, Electric cost, Water supply cost, Employee cost, Gas cost, operating cost.

For benefit, most of the revenue comes from renting. The following revenue are generated from the project-

Rent from shop.

Major revenue comes from renting shops for business. Market shop has always been attractive deal for renting.

Rent from office space.

Two floors will used as an office space. It also contributes in generating revenue.

Parking fee

There is a facility for parking. Fee will vary depending on car type.

Storage rent

There is a storage facility for market shop owners.

Rent from food corner.

One floor will be dedicated to the food corner.

The key assumptions taken during the financial analysis are:

- We assume the design value as 50 years and a construction period of 3 years. The construction cost has been equally distributed in these 3 years.
- A discount rate of 12% has been assumed for financial analysis.
- After the first 3 years of construction, the maintenance costs start, and a 15% increment considered in the maintenance cost every 10 years.
- For benefit, there is an increment of 20% every 5 years for the next 50 years. For the last year we added salvage value equal to 4% of the total construction cost.

From the financial analysis:

I. Financial Net Present Value (FNPV) = 231.08 crore

II. Financial Benefit Cost Ratio (FBCR) = 2.7

III. Financial Internal Rate of Return (FIRR) = 28.33%

6.2 Economic Analysis

The objective of this segment is to perform a thorough economic analysis, assessing both the costs and benefits associated with the construction of our project.

The direct and indirect cost and benefit components are:

Construction and Maintenance cost: the construction and maintenance cost has been converted to shadow price for economic analysis.

Job opportunity: Our shopping mall will create job opportunities for many people and they and their family will be economically benefited from this.

Increased property value: Our shopping mall will increase the value of property adjacent to it. It has been considered in economic analysis.

Economic benefit from Green Park: Economic benefit from Green Park has also been considered in economic analysis.

Fire safety enhancement: Our building will be constructed with proper and modern fire safety measures which will reduce the risk of any fire accident.

Earthquake Risk Reduction: The building will be constructed following BNBC code and proper earthquake safety measures will be maintained to minimize the damage.

Proper waste management system: Proper waste management system will be constructed which will reduce the possibility of disease and other environmental issues.

Fuel cost reduction: We have built parking facilities which will reduce the fuel cost of the cars parked in parking facilities.

Assumptions:

- We assume the design value as 50 years and a construction period of 3 years. The construction cost has been equally distributed in these 3 years.
- We have multiplied the market price by 0.8 to convert it into shadow value where adjustment is necessary.
- After the first 3 years of construction, for the maintenance costs and benefits a 15% increment is considered every 10 years.
- A discount rate of 12% has been assumed for economic analysis.

The following result has been found from economic analysis:

I. Economic Net Present Value (ENPV): 130.33 crore
II. Economic Benefit Cost Ratio (EBCR): 1.83
III. Economic Internal Rate of Return (EIRR): 24%

Section 7: Human Resources and Administrative Support Analysis

(a) Types of Managerial and Skilled Workforces Needed During Implementation and Operational Phases:

A varied workforce may be needed by the Dhaka North City Corporation throughout the project's implementation phase. Project managers, civil engineers, architects, builders, procurement and logistics experts, communicators, and community involvement specialists might all fall under this category.

In the operational phase, staffing needs may shift towards maintenance and facility management. Skilled personnel such as the shopping complex managers, security staff, maintenance crews, and administrative staff would be essential.

(b) Ability of the Project Entity to Provide Workforces:

Assessing the Dhaka North City Corporation's ability to provide the required workforces involves an analysis of existing staff capabilities. If there are gaps, the corporation might need to consider recruitment, training programs, or collaborations with external agencies possessing the necessary expertise.

(c) Institutional Capacity for Retaining Project Output:

Ensuring the institutional capacity of the Dhaka North City Corporation involves evaluating both financial and technical aspects. If there are deficiencies, strategies might include seeking additional funding, forging partnerships, or investing in training programs for existing staff.

(d) Equipped with Skilled and Experienced Workforces:

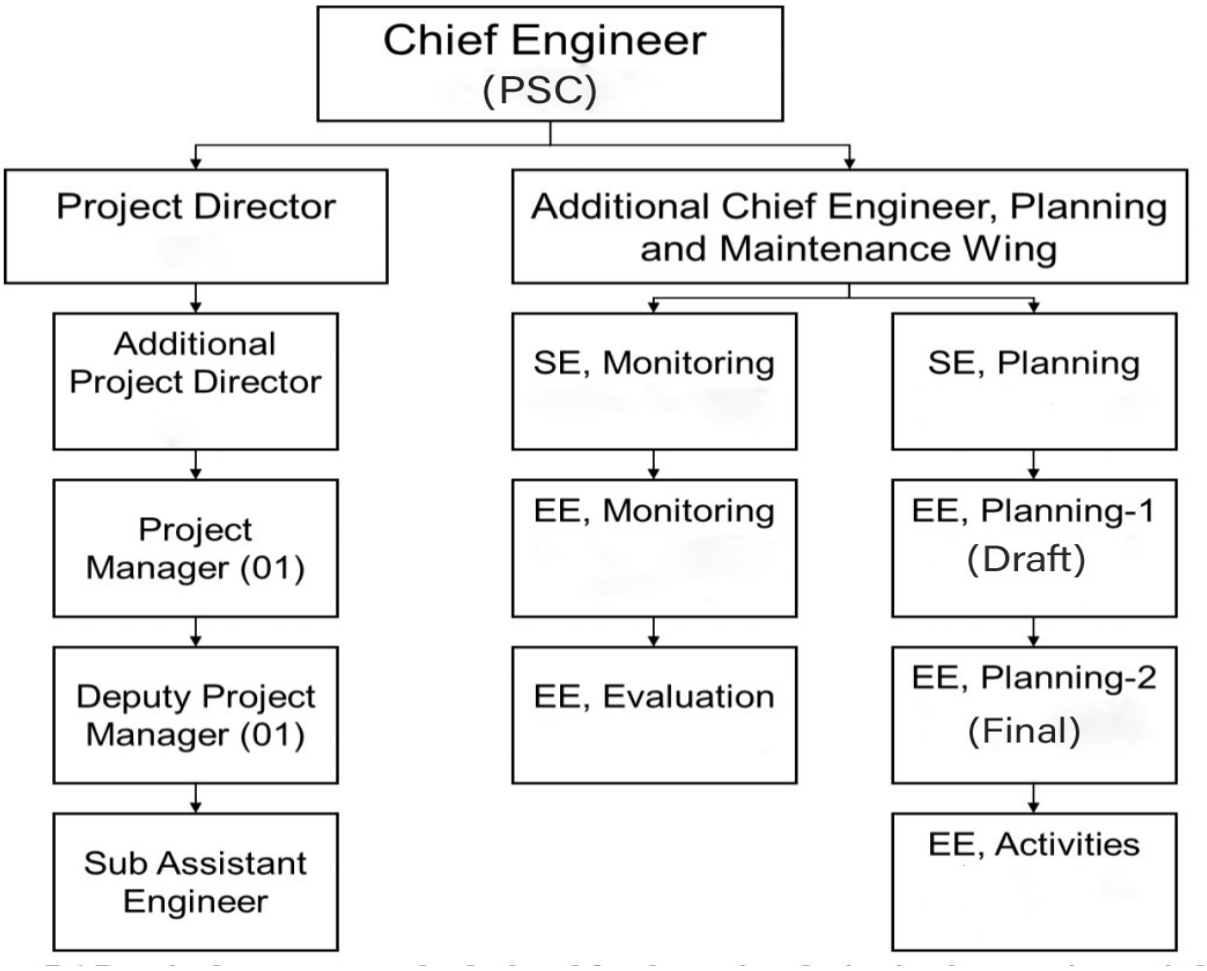
A skills assessment within the Dhaka North City Corporation is crucial to determine if the entity is equipped with the necessary personnel to operate the Multi-purpose Shopping Complex after completion. If deficiencies exist, addressing them through training initiatives, hiring, or outsourcing specific functions may be necessary. For supervision of the project implementation, there will be a Project Steering Committee (PSC) for overall guidance and a Project Implementation Unit (PIU) for overall supervision. The PSC may be chaired by the Chief Engineer and the Project Director (PD) may be the member secretary. There may be representation from the Planning Commission, IMED and Upazila Parishad in the PSC. The PIU may be headed by the PD.

(e) Adequate Funding for Operational Expenditure:

Reviewing the recurring budget is essential to ensure the Dhaka North City Corporation has sufficient funds for the operational expenditure of the Multi-purpose Mohammadpur Shopping Complex. If there is a shortfall, the corporation might need to explore alternative funding sources or revise the budget.

(f) Timing of Project Consistent with Organizational Capacity:

The Dhaka North City Corporation should assess the project timeline in comparison to its organizational capacity. If there were challenges in past project implementations, adjustments to the timeline, resource allocation, or process optimization may be necessary.



Required skilled manpower to be deployed for the project

Section 8: Institutional and Legal Analysis

(a) The Development of a Multi-storied Shopping Complex at Mohammadpur Krishi Market project aligns seamlessly with the legal boundaries and the mandated responsibilities of the Dhaka North City Corporation, ensuring compliance with all relevant laws and regulations governing urban development and transportation in Bangladesh.

(b) Efforts have been made to optimize the utilization of resources within the Dhaka North City Corporation. A comprehensive review indicates efficient use of existing capabilities and physical facilities to support the successful implementation of the Multipurpose Shopping Complex Development project.

(c) No significant adjustments or reforms are currently deemed necessary in the policy and institutional setup. Existing policies and structures sufficiently support the goals and requirements of the project.

(d) Minor adjustments, mainly procedural in nature, may be required to ensure seamless project implementation. These adjustments will be made in strict accordance with legal frameworks and organizational policies.

(e) The Dhaka North City Corporation possesses the necessary skills and capacity to effectively manage and implement the Multipurpose Shopping Complex Development project. Any minor skill gaps will be addressed through targeted training programs to enhance workforce capabilities.

(f) To encourage the timely and cost-compliant completion of projects, incentive systems have been put in place. Clear contractual terms and performance standards are also in place, along with consequences for any delays or overspending.

(g) No critical governance issues that may significantly impact project implementation have been identified. Governance structures are robust, ensuring transparency, accountability, and effective stakeholder engagement throughout the project lifecycle.

(h) While potential cross-cutting issues, such as environmental considerations, have been acknowledged, a proactive mitigation strategy is in place. This includes the implementation of environmental impact assessments and community engagement plans to address and minimize potential challenges.

Section 9: Risk and Sensitivity Analysis

The flow of costs and benefits throughout the project life is uncertain. Given that uncertainty, considerations have to be given to the costs that risks imply. The objective of this module is to simulate various scenarios and generate guidance on how to reduce the risk exposure through relevant contractual clauses.

1. The major risks to the project are listed below:

(i) **Inclement weather**: Every year heavy rainfall in Dhaka city occurs. Construction operation is disrupted under heavy rainfall. The construction phase is also vulnerable to high wind speed.

(ii) **Natural calamities**: The most probable natural calamities in the area is earthquake. There have been mild earthquake instances in the area. There is a high risk of earthquake. According to the expert, up to 300,000 people may die if a magnitude 7 earthquake strikes Dhaka right now(dhakatribune.com)

(iii) **Pandemic and other health hazards:** Covid 19 like pandemic or other health hazards may reduce the number of customers. These will badly effect the market's selling rate and profit. These health hazards may also warrant different health safety measures.

(iv) **Fire and other human induced hazards**: These hazards may cause damage to the building.

(v) **Malfunction of equipment**: Electrical or mechanical malfunction of equipment may cause temporary shutdown of the shopping complex.

(vi) **Socio-political unrest**: Social or political unrest in the area may reduce the number of visitors or temporary shutdown of the facility.

(vii) **Economic instability**: Due to high inflation the maintenance operation may be less cost effective.

2. There may be different degrees of impact if the risk events are materialized. The impacts are listed below:

(i) **Prolonged shutdown or lockdown**: Due to natural calamities, pandemic, etc. there may be prolonged shutdown or lockdown of the shopping complex. This may significantly reduce the revenue of the project.

(ii) **Temporary shutdown or lockdown**: Due to inclement weather, minor natural calamities, pandemic or other health hazards, human induced hazards, malfunction of equipment, socio-political unrest etc. there may be temporary unplanned shutdown or lockdown of the multi-purpose shopping complex. This may reduce the revenue to some extent.

(iii) **Evacuation**: Sudden weather conditions, natural disaster, fire or other human induced hazards, malfunction of equipment etc. may cause evacuation of the facility and emergency response to tackle the situation.

(iv) **Difficulty in accessing the facility**: Due to calamities like earthquake damage in nearby areas or socio-political unrest accessing the multi-purpose shopping complex may become difficult. This type of situation will reduce the demand and revenue of the project.

(v) **Reduction in number of visitors**: Demand of the shopping complex may reduce due to pandemic, socio-political unrest or economic instability or online marketing.

3. Mitigation measures for different risk events are different. Possible mitigation measures are listed below:

(i) **Following weather forecasts**: To mitigate problems due to inclement weather, the weather forecasts should be closely monitored, and necessary steps should be taken.

(ii) **Disaster resilient design**: The effects of different natural disasters should be considered during the design phase.

(iii) **Emergency response**: There has to be a proper planning for emergency response to mitigate the effects of natural disasters. A detail discussion on this topic has been presented in Section 5.Health safety measures: Proper health and safety measures should be practiced in the facility.

(iv) **Maintenance contract with the manufacturer**: An annual maintenance contract should be maintained with the manufacturer so that risk is shared.

(v) **Alternate source of critical parts**: Alternate source of critical parts should be identified so that in case the manufacturer fails to deliver the parts, the critical parts can be availed from an alternate source.

(vi) **Capacity building of local industry**: There should be efforts to improve capacity of the local industry so that they can support maintenance of the building with a reasonable cost.

(vii) **Dialogue with local communities**: The authority should engage themselves in regular dialogue with local communities so that any grievance of the community can be timely addressed.

4. No legal or regulatory obligations can increase the cost or lower the benefit.

Sensitivity analysis

Table : Financial Sensitivity Analysis

Case	Sensitivity Scenario	NPV (BDT in crore)	IRR(%)	BCR	Comment
1	Base cost and base benefits	-14.79	10.86	0.89	< Discount Rate(12%)
2	Decrease costs by 20% and base benefits	11.66	13.09	1.11	> Discount Rate(12%)
3	Base Cost and increase benefits by 20%	8.70	12.65	1.07	> Discount Rate(12%)
4	Decrease capital costs by 10% and increase benefits by 10%	10.18	12.85	1.09	> Discount Rate(12%)

Table : Economic Sensitivity Analysis Results

Case	Sensitivity Scenario	NPV(BDT in crores)	EIRR(%)	Comment
1	Base cost and base benefits	102.42	23	> Discount Rate(12%)

2	Increase costs by 30% and base benefits	70.72	18	> Discount Rate(12%)
3	Base Cost and decrease benefits by 30%	38.89	16	> Discount Rate(12%)
4	Increase capital costs by 20% and Decrease benefits by 20%	39.67	16	> Discount Rate(12%)

Section 10: Analysis of Alternatives

Making decision for the type of structure of the building:

Criteria	Beam-supported Slab	Flat Slab	Steel Structure
Required Materials	Typically consists of concrete beams and a concrete slab supported by columns. Steel reinforcement is used.	Consists of a flat, reinforced concrete slab without beams supported by columns. It may include drop panels or column heads.	Typically consists of concrete slab cast on steel deck supported by primary and secondary beams which rest on columns, all of which are made of steel sections.
Costing	Generally, it is more cost-effective for smaller spans but may require more materials and labor for the casting of beams.	Moderately expensive due to the additional reinforcement required for load distribution.	Generally, larger spans tend to result in higher costs due to the increased material quantities, fabrication complexity, transportation expenses
Construction Technology	Requires formwork for beams and slab, followed by concrete pouring and curing.	Similar to a beamsupported slab but without beams, may require additional reinforcement.	Prefabricated steel beams and columns are connected using end plates via welding or using bolts.

Structural Integrity	The presence of beams provides added structural integrity	Proper detailing of reinforcement and connections is essential to ensure structural integrity, especially at column slab junctions.	The structural integrity of steel structures primarily comes from the inherent properties of steel, such as its high strength-to-weight ratio, ductility, and corrosion resistance. Additionally, structural integrity relies on the proper design, fabrication, and construction methods employed.
Strength	Beam-supported slabs are typically strong due to the presence of beams which help distribute loads effectively to the supporting columns. The beams add stiffness and reduce deflection.	Flat slabs are strong but require careful detailing to ensure adequate load distribution and structural integrity, especially at column connections.	The strength of a steel structure depends on factors such as the quality of the steel material, the design of structural members, the efficiency of connections, and the overall structural system
Fire Resistance	Beam-supported slabs typically offer better fire resistance compared to flat plate slabs due to their higher thermal mass, partial shielding of reinforcement, and load redistribution mechanism.	In contrast, flat plates have no such shielding, which may result in quicker deterioration of the reinforcement. flat plates may exhibit a more sudden and widespread failure due to the lack of beam support.	The members of steel structures get directly exposed to fire. So the temperature of steel rises quickly and load carrying capacity decreases significantly at high temperatures.
Earthquake Resistance	It is more resistant to earthquakes as it is more flexible than a flat slab system.	It is less resistant to earthquakes as it is less flexible than the slab beam system. The formation of hinges at the column slab joint could be catastrophic.	Steel structures exhibit inherent earthquake resistance due to the material's high strength, ductility, and flexibility.

Architectural Considerations	Beam-supported slabs offer moderate aesthetic freedom and floor-to-ceiling heights, along with limited structural expression and moderate spatial planning efficiency.	Flat plates provide high aesthetic freedom, shorter floor-to-ceiling heights, moderate structural expression, efficient spatial planning, and high construction efficiency at low cost.	Flat plates provide high aesthetic freedom, shorter floor-to-ceiling heights, moderate structural expression, efficient spatial planning, and high construction efficiency at low cost.

Decision: RCC building will be constructed.

Making decision for the foundation of the building

Criteria	**Raft foundation (main design)**	**Shallow foundation (alternative design)**	**Decision**
Load Distribution	Distributes the weight of the building evenly over a large area, providing stability and preventing settlement.	Spreads the building load over a wider soil area, reducing the risk of soilbearing capacity failure and settling.	Raft is foundation preferred.
Use Cases	Suitable for structures requiring heavy load-bearing capacity, such as high-rise buildings, industrial facilities, or large commercial developments.	Commonly used for residential buildings, smaller commercial structures, and light industrial buildings with relatively lighter loads.	Raft is foundation preferred.
Soil Conditions	Can accommodate various soil types and conditions, including poor soil conditions or areas with high water tables, with appropriate design and reinforcement.	Ideal for stable soil conditions with sufficient bearing capacity to support the building load without excessive settlement.	Raft foundation is preferred.

Cost	Generally, more expensive to construct due to the use of reinforced concrete and the need for specialized engineering and construction techniques.	Typically, less expensive compared to deep foundations, as they require less excavation and material, and are simpler to construct.	A shallow foundation is preferred.
Settlement	Resists partial settlement, allowable settlement of soil is relatively large.	Partial settlement of soil may occur, but allowable settlement is relatively small.	Raft foundation is preferred.

Decision: Raft foundation will be constructed.

Section 11: Conclusion and Recommendation

In conclusion, the feasibility analysis of the proposed project has brought to light various significant factors that need careful consideration. The identified risks, encompassing inclement weather, natural calamities, health hazards, and socio-political unrest, underscore the importance of robust risk mitigation strategies. The simulation of scenarios has allowed for a comprehensive understanding of potential impacts on costs and benefits throughout the project life.

Risk Assessment:

Identified major risks include inclement weather, natural calamities, health hazards, fire, equipment malfunction, socio-political unrest and economic instability.

Potential impacts range from prolonged shutdowns to reduced visitor numbers, necessitating a strategic approach to risk management.

Mitigation Measures:

Mitigation strategies involve weather monitoring, disaster-resilient design, emergency response planning, health safety measures, equipment maintenance and community engagement.

Establishing alternate sources for critical parts and fostering the capacity of local industries are integral components of the risk mitigation plan.

Financial and Economic Models:

Sensitivity analysis reveals the impact of varying growth rates, discount rates, fees, rents and land prices on financial and economic models.

Alternative/Options Analysis:

Comparison between beam-supported slab and flat plate systems, along with mat and pile foundations, indicates that the beam-supported slab with a mat foundation is more favorable.

Factors influencing the decision include cost, construction time, technological requirements, performance under seismic activity and social acceptability.

Financial Evaluation:

The financial models indicate varying Net Present Values (NPV), Benefit-Cost Ratios (BCR), and Internal Rates of Return (IRR) based on growth rates, discount rates, fees, hotel rents and land prices.

Recommendation:

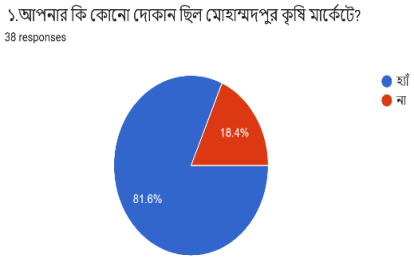
- The beam-supported slab system is recommended over the flat plate system due to its superior performance in earthquake resistance.
- Mat foundation is favored over pile foundation considering factors such as cost, construction time, technological requirements and social acceptability.
- Considering the comprehensive analysis, it is recommended to proceed with the project, implementing the suggested risk mitigation measures. The beam-supported slab with a mat foundation is the optimal design choice, ensuring structural integrity, cost-effectiveness and timely completion. Regular monitoring and adaptation to economic variables are advised to enhance the project's long-term viability.

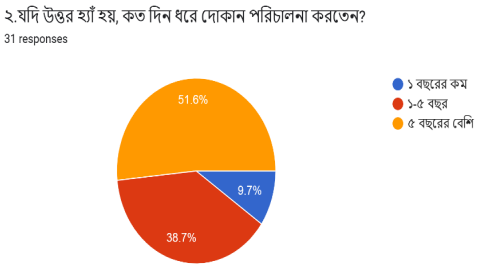
Section 12: Annexes

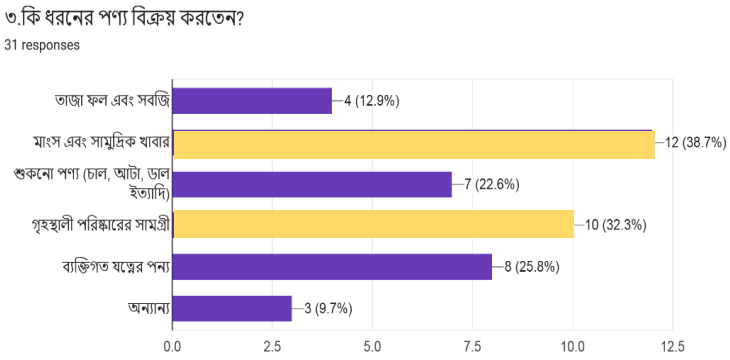
12.1 Questionnaire Survey for Local Community

Market Demand Survey Results:

Survey for Seller:

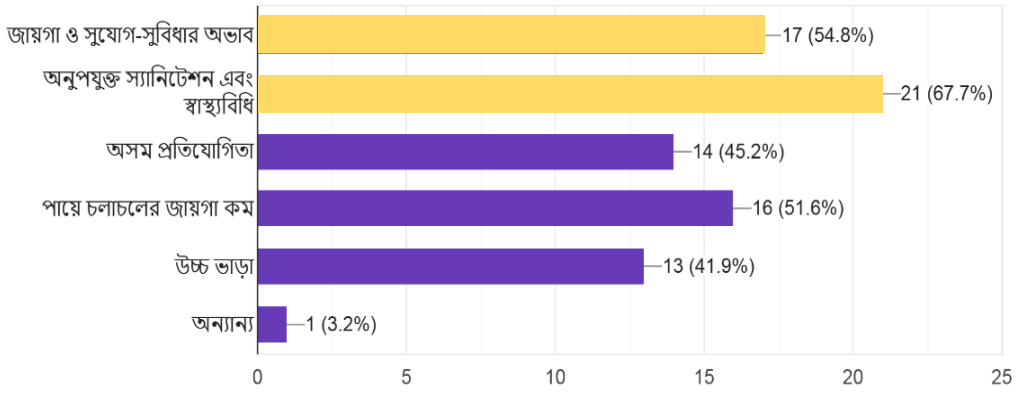






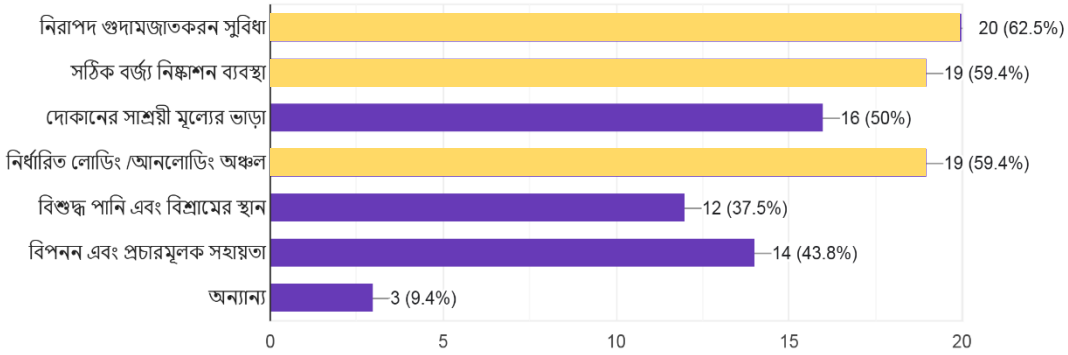
৪.কি ধরনের সমস্যা মোকাবিলা করতে হয়েছিল দোকানে থাকাকালীন?

31 responses



৫.নতুন বাজারের কোন বৈশিষ্ট্য বা সুবিধাগুলো আপনার জন্য বেশি সহায়ক হবে?

32 responses

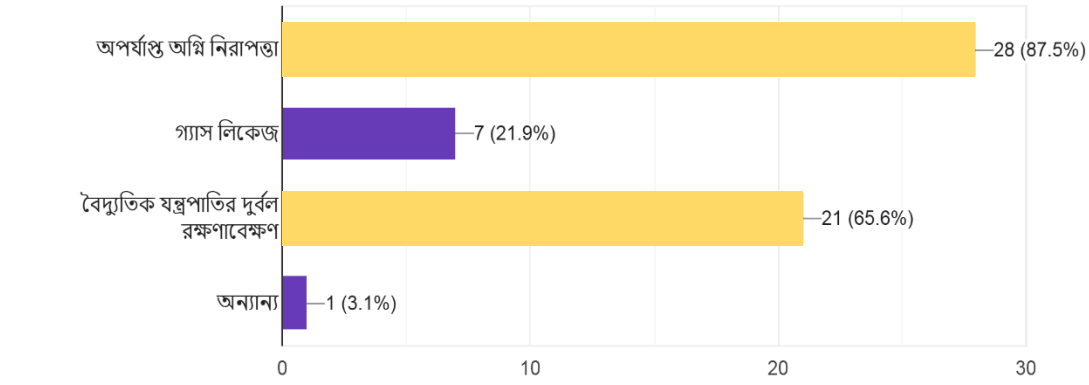


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৭.পূর্ববর্তী অগ্নিকাণ্ডের জন্য কোন প্রাদুর্ভাবের ভুমিকা সবচেয়ে বেশি বলে আপনি মনে করেন?

32 responses



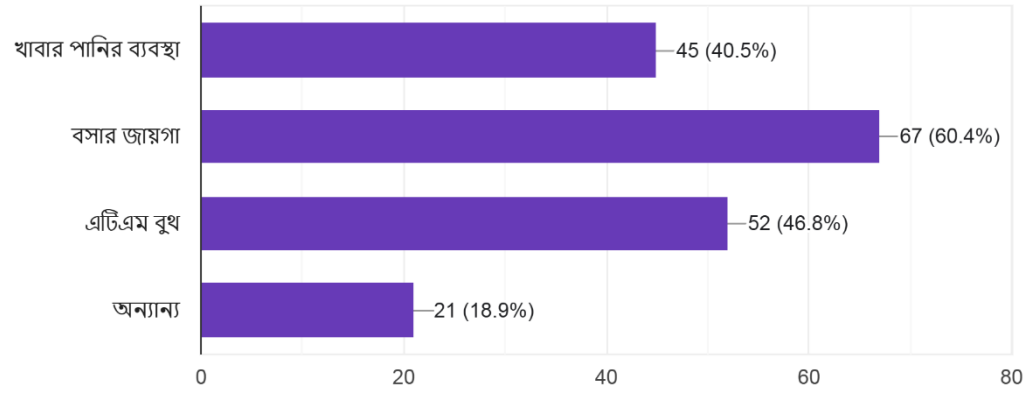
তাজা এ

Survey for Customer:

১. বর্তমানে আপনি শাক সবজি এবং যাবতীয় রান্নার জিনিসপত্র কোথায় থেকে কিনছেন?

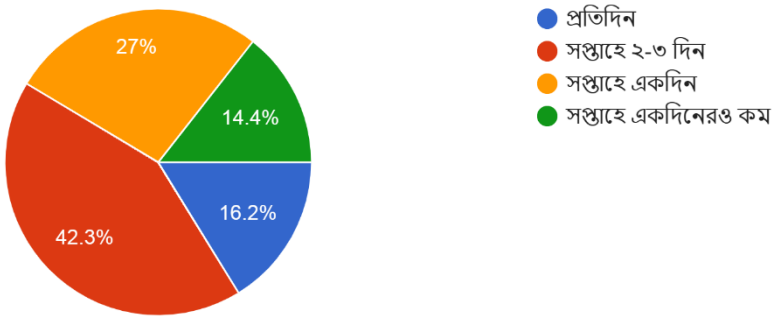
২. কৃষি মার্কেটে নতুন শপিং কমপ্লেক্স বানানো হলে কি কি সুবিধা থাকলে ভালো হয় বলে আপনি মনে করেন?

111 responses



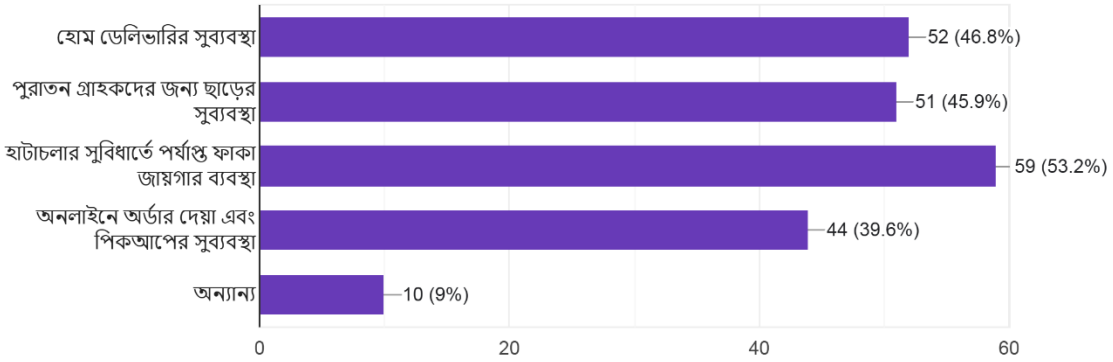
৩. সপ্তাহে কতদিন আপনি শাক সবজি এবং এবং রান্নার জিনিসপত্র কেনাকাটা করেন ?

110 responses



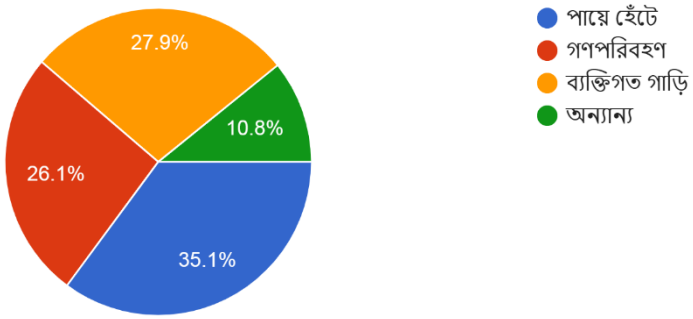
৪. অতিরিক্ত কোন সেবাটি আপনার কেনাকাটার অভিজ্ঞতাকে বৃদ্ধি করবে বলে আপনি মনে করেন?

111 responses



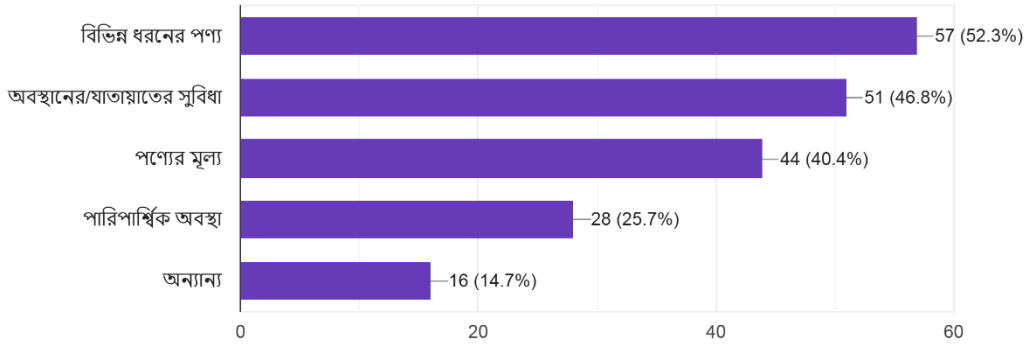
৫. আপনি সাধারণত কিভাবে বাজার করে থাকেন ?

110 responses



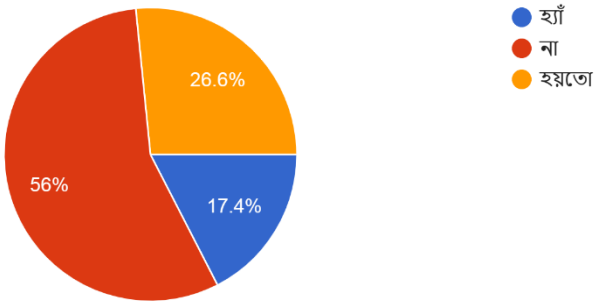
৭. পুর্বের বাজারের কোন দিকটি আপনি সবচেয়ে পছন্দ করতেন ?

109 responses



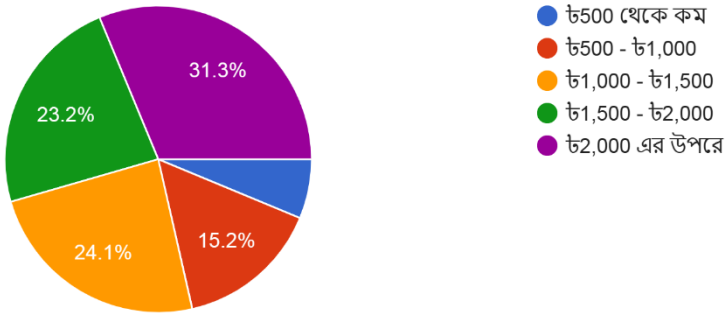
৬. আপনি কি সামাজিক সমাবেশ বা ইভেন্টের জন্য বাজারের মধ্যে কিছু জায়গা রাখতে পছন্দ করবেন?

108 responses



৮. আপনি প্রতি সপ্তাহে মুদি এবং সবজির জন্য কতটা ব্যয় করে থাকেন ?

111 responses



12.2 Load Calculation and Design

Wind Load Analysis:

Velocity Pressure Exposure Coefficients, K_h	0.974
Velocity pressure, q_h	2.191
External Pressure Coefficient (Windward), C_{PWX}	0.8
External Pressure Coefficient (Leeward), C_{PLX}	0.39
External Pressure Coefficient (Windward), C_{PWY}	0.8
External Pressure Coefficient (Leeward), C_{PLY}	0.5
Building Period, T (Seconds)	0.964
Building Flexible?	NO
Gust Factor, G_X	0.826
Gust Factor, G_Y	0.81

Length in X Direction, L_X	73.135	meter
Length in Y Direction, L_Y	45.72	meter
Total Height, h	28.956	meter
Story Height	4.572	meter
No. of Stories	8	
Exposure	A	
Location	Dhaka	
Wind Speed	65.7	m/s
Importance Factor, I	1	
Strecture Type	Concrete moment-resisting frames	
Damping Ration, b	5	

Tropograpgic Factor, k_{zt} = 1

Wind Directionality Factor, k_d = 0.85

Floor	Height Above Ground Level, Z (m)	Velocity Pressure Exposure Coefficient, K_z	Velocity Pressure, q_z (kN/m^2)	Design Pressure, Px (Windward) (kN/m^2)	Design Pressure, Px (Leeward) (kN/m^2)	Design Pressure, Py (Windward) (kN/m^2)	Design Pressure, Py (Leeward) (kN/m^2)	Projected Area, Ax (m^2)	Projected Area, Ay (m^2)	Design Wind Force, Fx (Windward) (kN)	Design Wind Force, Fx (Leeward) (kN)	Design Wind Force, Fx (Total) (kN)	Design Wind Force, Fy (Windward) (kN)	Design Wind Force, Fy (Leeward) (kN)	Design Wind Force, Fy (Total) (kN)
Ground Level	0	0.575	1.293	0.854	0.687	0.838	0.904	0	0	0	0	0	0	0	0
F1	4.572	0.575	1.293	0.854	0.687	0.838	0.904	209.032	334.374	179	144	322	280	303	583
F2	9.144	0.701	1.576	1.041	0.687	1.021	0.904	209.032	334.374	218	144	361	342	303	644
F3	13.716	0.787	1.77	1.168	0.687	1.146	0.904	209.032	334.374	245	144	388	384	303	686
F4	18.288	0.855	1.921	1.268	0.687	1.244	0.904	209.032	334.374	266	144	409	416	303	719
F5	22.86	0.911	2.048	1.352	0.687	1.326	0.904	209.032	334.374	283	144	427	444	303	746
F6	27.432	0.959	2.157	1.424	0.687	1.397	0.904	209.032	334.374	298	144	442	468	303	770
F7	32.004	1.003	2.254	1.488	0.687	1.46	0.904	209.032	334.374	311	144	455	489	303	791
F8	36.576	1.042	2.342	1.546	0.687	1.517	0.904	104.516	167.187	162	72	234	254	152	405

Seismic Load Analysis:

Height of Building (m)	28.956
Strecture Type	Concrete moment-resisting frames
Site Class	SC
Viscous Damping Ratio, ζ	5
Response Reduction Factor, R	5
Occupancy Category	II
Seismic Zone	2, Z=0.2

C_t	0.0466	[From Table 6.2.20 of BNBC 2020]
m	0.9	[From Table 6.2.20 of BNBC 2020]
S	1.15	[From Table 6.2.16 of BNBC 2020]
T_B	0.2	[From Table 6.2.16 of BNBC 2020]
T_C	0.6	[From Table 6.2.16 of BNBC 2020]
T_D	2	[From Table 6.2.16 of BNBC 2020]
Damping Correction Factor,η	1	[η= √(10/(5+ζ)) ≥ 0.55]
Building Period, T (Seconds)	0.964	[$T = C_t(h_n)^m$]
Normalized acceleration response spectrum, C_S	1.79	[2.5 S η (TC / T)]
Importance Factor,I	1	[From Table 6.2.17 of BNBC 2020]
Seismic Zone Coefficient, Z	0.2	[From Table 6.2.14 of BNBC 2020]
Response Reduction Factor, R	8	

$$S_a = \frac{2}{3}\frac{ZIC_s}{R}$$

Lateral Seismic Force Coefficient, Sa	0.0477

V= Sa*W = 0.0477*W, 4.77% of Seismic W

Column=(24X24)/144 *95*88*.15=5016 kip

Beam=(12X24)/144 *3612*8*.15=57792 kip

Slab Thickness=6”

Slab SW: (6/12)*150=75 psf

PW : 60 psf

FF : 30 psf

Total DL from slab= (75+60+30) x (240X150) /1000 = 5940 kip

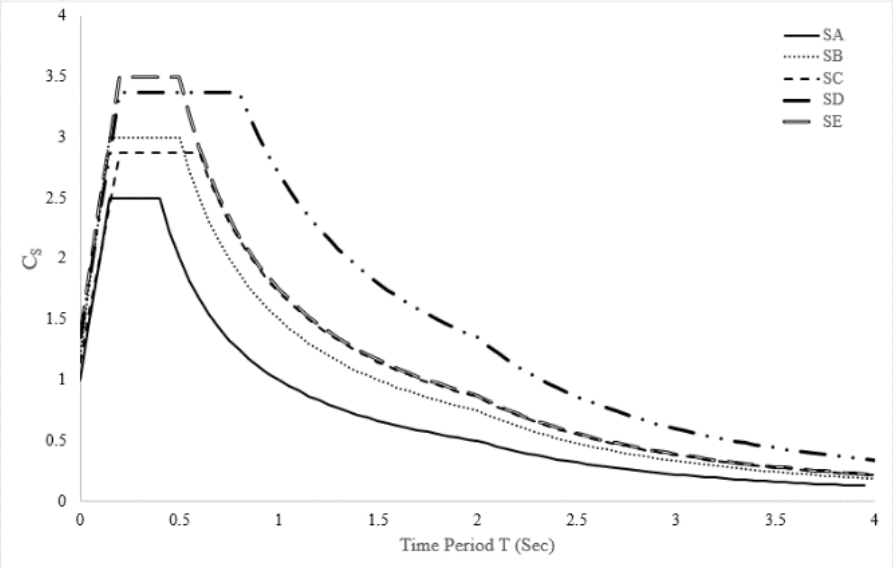
Total LL from slab= 40 x 240*150/ 1000= 1440 kip

Total dead load from building = 5940*8+5016+57792=110328 kip

Total live load from building=8*1440=11520 kip

Total seismic load, W=110328+0.25*11520=113208 kip

V= Sa*W = 0.0477*113208=5400 kip



	C_S				
T	SA	SB	SC	SD	SE
0	1	1.200	1.150	1.350	1.400
0.05	1.5	1.800	1.581	1.856	1.925
0.1	2	2.400	2.013	2.363	2.450
0.15	2.5	3.000	2.444	2.869	2.975
0.2	2.5	3.000	2.875	3.375	3.500
0.25	2.5	3.000	2.875	3.375	3.500
0.3	2.5	3.000	2.875	3.375	3.500
0.35	2.5	3.000	2.875	3.375	3.500
0.4	2.5	3.000	2.875	3.375	3.500
0.45	2.222	3.000	2.875	3.375	3.500
0.5	2.000	3.000	2.875	3.375	3.500
0.55	1.818	2.727	2.875	3.375	3.182
0.6	1.667	2.500	2.875	3.375	2.917
0.65	1.538	2.308	2.654	3.375	2.692
0.7	1.429	2.143	2.464	3.375	2.500
0.75	1.333	2.000	2.300	3.375	2.333
0.8	1.250	1.875	2.156	3.375	2.188
0.85	1.176	1.765	2.029	3.176	2.059
0.9	1.111	1.667	1.917	3.000	1.944
0.95	1.053	1.579	1.816	2.842	1.842
1	1.000	1.500	1.725	2.700	1.750
1.05	0.952	1.429	1.643	2.571	1.667
1.1	0.909	1.364	1.568	2.455	1.591

Lift Area and Number Determination:

4.3.1.6 Handling capacity and interval

(a) The handling capacity, for incoming/up-peak passenger, shall be calculated by the following formula:

$$H = \frac{300 \times Q \times N \times 100}{P \times T}$$

Where,

H = passenger handling capacity of the lift system during five minute peak period, expressed as the percentage of the estimated total population handled.

Q = average number of passengers carried in each car per trip,

N = number of lifts in the system,

P = total population to be handled during peak period (it is related to the area for which a particular bank of lift serves)

T = average round trip time in seconds, that is, the average time required by each lift in taking one full load of passengers from ground floor discharging them in various upper floors and coming back to ground floor for taking fresh passengers for the next trip;

Interval, T= 40-50 sec. (Category Shops and Stores in BNBC Table 8.4.7)

Handling Capacity, H = 5% (Category Shops and Stores in BNBC Table 8.4.7)

Population factor = $2m^2$ (Category Shops and Stores in BNBC Table 8.4.8)

Area = 3210 m

P = 2x 3210 = 6420

Rise -> 90 ft -> (2 -30 m)

Minimum Cargo speed = 1m/s.

$$H = \frac{300QN}{PT} * 100$$

Let, N=4 (No. of lift.) $$5 = \frac{300Q*4}{6420 * 40} * 100$$

∴ Q = 10.7 ≈ 11 Person/Lift.

Max available lift area = $1.9\ m^2$ (From BNBC Table 8.4.1)

Number of lift = 4

Available lift area per lift = $1.9\ m^2$

Floor Space Usage:

Basement 1 : Car parking

Basement 2 : Car parking

Basement 3 : Car parking

Level-01: Wholesale vegetable market and market space

Level-02: Groceries shops, clothing store shops, market space

Level-03: Super shop, Food court and prayer room

Level 04 to 08: Office space

Parking and Basement Calculation:

Level 1,2 - shop || Level-3 Shop + Restaurant || Level 4 to 8 = Office

FOR SHOPS:

1 car parking for 200m^2 (2160$ft^{2)}$ Gross area: (From Table 3.F.3)

Level-01 = 240x150 = 36000 ft^2
Level-02 = 240x150= 36000 ft^2
Level-03 = 240x84= 36000 ft^2
TOTAL = 92160ft^2
Car parking need for shops = 92160/2160

= 42.67

= nos. 43

FOODCOURT/RESTURANTS:

1 car parking per 100m^2 (1080ft^2) gross area: (From Table 3.F.3)

Level-03 = 240x66=15840ft^2

Car parking need for food court = 15840/1080

= 14.6

= nos. 15

OFFICE:

1 car parking for 200m^2 (2160$ft^{2)}$ Gross area):

Level 4 to 8 =5x(240x150) =180000ft^2

Car parking need for office= 180000/216 = 83.33 = nos.84

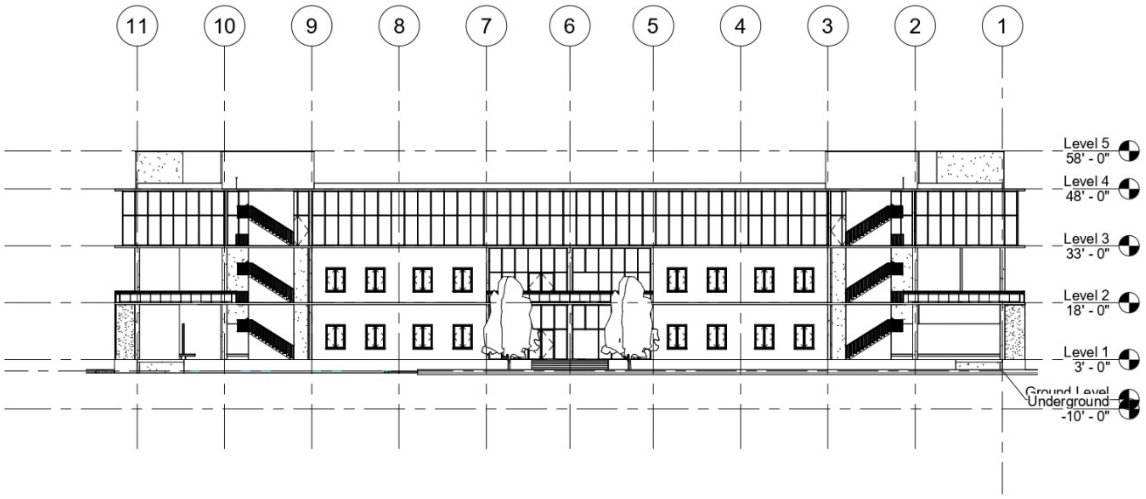
Total car parking needed= 43+15+84= nos.142

Per basement capacity = 48

Providing 3 basement and front parking = 3x50 =150

Basement Provided = 3

Elevation View:



Preliminary Design:

Load Calculation

Dead Load:

Let, Slab Thickness = 6”

Floor Finish = 25 psf

Partition Wall = 50 psf

Live Load:

Office Buildings or Restaurants
= 100.25 psf

Stores = 100.25 psf

Stairs = 100 psf

Beams

Let, b_w = 12”

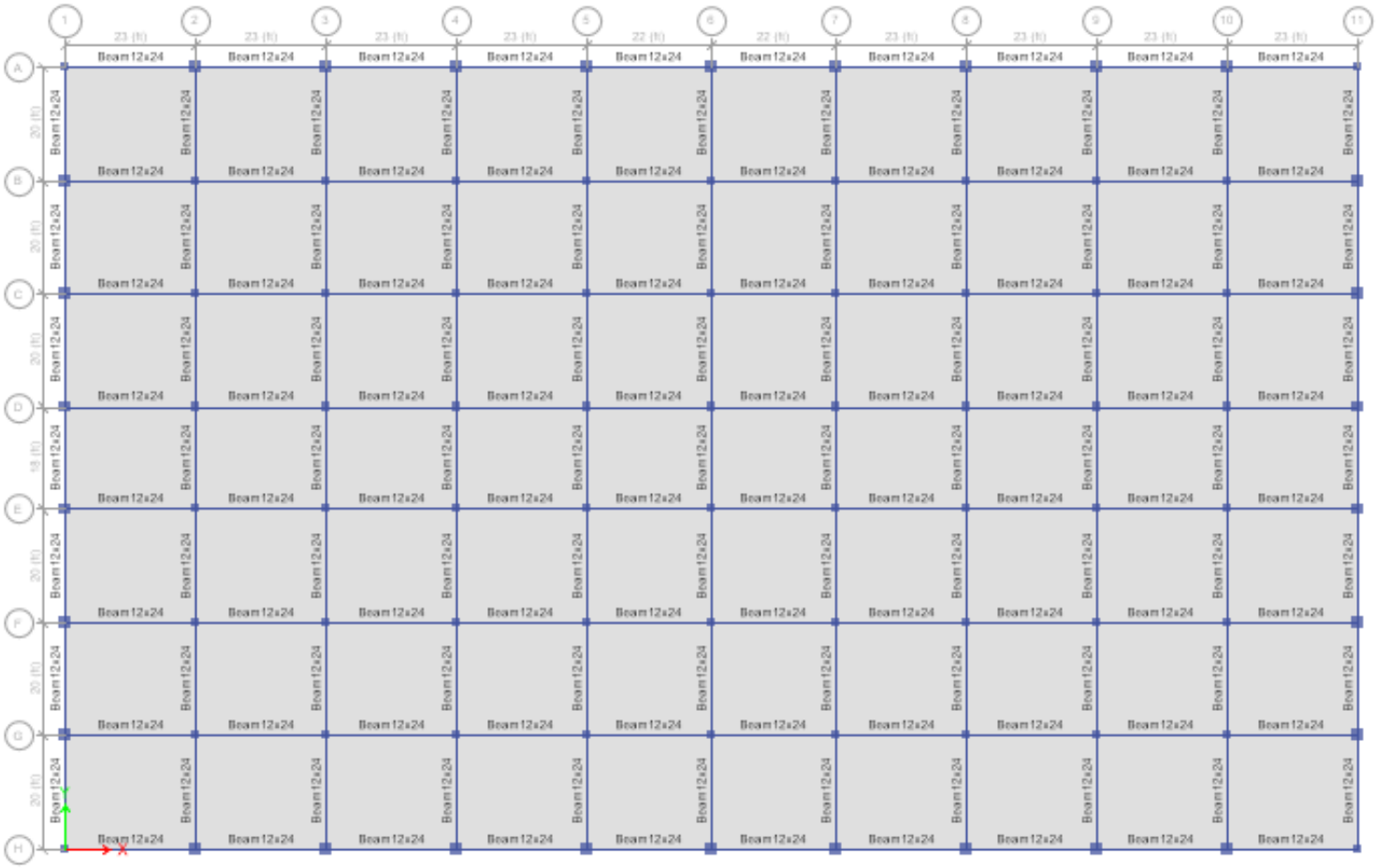
For Floor Beams, maximum depth for both end continuous beams = 30 * 12 / 21 = 17.14” = 18”
(rounded to upper 2”) which is equal to 3*slab thickness (OK)

maximum depth for one end continuous beams = 30*12 /18.5 = 24” > 3*slab thickness (OK)

We are taking **12” X 24” Floor Beams**.

For Stair Beams, maximum depth = 10*12/21= 5.71” < 18”

We are taking **12” X 18” Stair Beams**.



Beams Layout

Columns

Corner Columns

Wu = 1.2*(6/12*150 + 25 + 50)*11 +1.6*100.25*11 psf = 3744.44 psf

Area = 20*23/4 sft

So, Pu = 3744.44 *20*23/4 = 430.61 kip

Assuming 1% reinforcement, P u = $\alpha \emptyset Ag[0.85fc'(1-\rho)+\rho fy]$

Here, fc' = 4 ksi and fy = 60 ksi (assumed)

Ag = 208.8 sq.inch

Assuming square column, a 14.45” x 14.45” column is obtained.

We are taking 18” x 18” columns.

Edge Columns

Wu = 1.2*(6/12*150 + 25 + 50)*11 +1.6*100.25*11 psf = 3744.44 psf

Area = 20*23/2 sft

So, Pu = 3744.44 *20*23/2 = 861.22 kip

Assuming 1% reinforcement, P u = $\alpha \emptyset Ag[0.85fc' (1 - \rho) + \rho fy]$

Here, fc' = 4 ksi and fy = 60 ksi (assumed)

Ag = 417.59 sq.inch

Assuming square column, a 20.42" x 20.42" column is obtained.

We are taking 24" x 24" columns.

Center Columns

Wu = 1.2*(6/12*150 + 25 + 50)*11 +1.6*100.25*11 psf = 3744.44 psf

Area = 20*23 sft

So, Pu = 3744.44 *20*23 = 1722.44 kip

Assuming 1% reinforcement, P u = $\alpha \emptyset Ag[0.85fc' (1 - \rho) + \rho fy]$

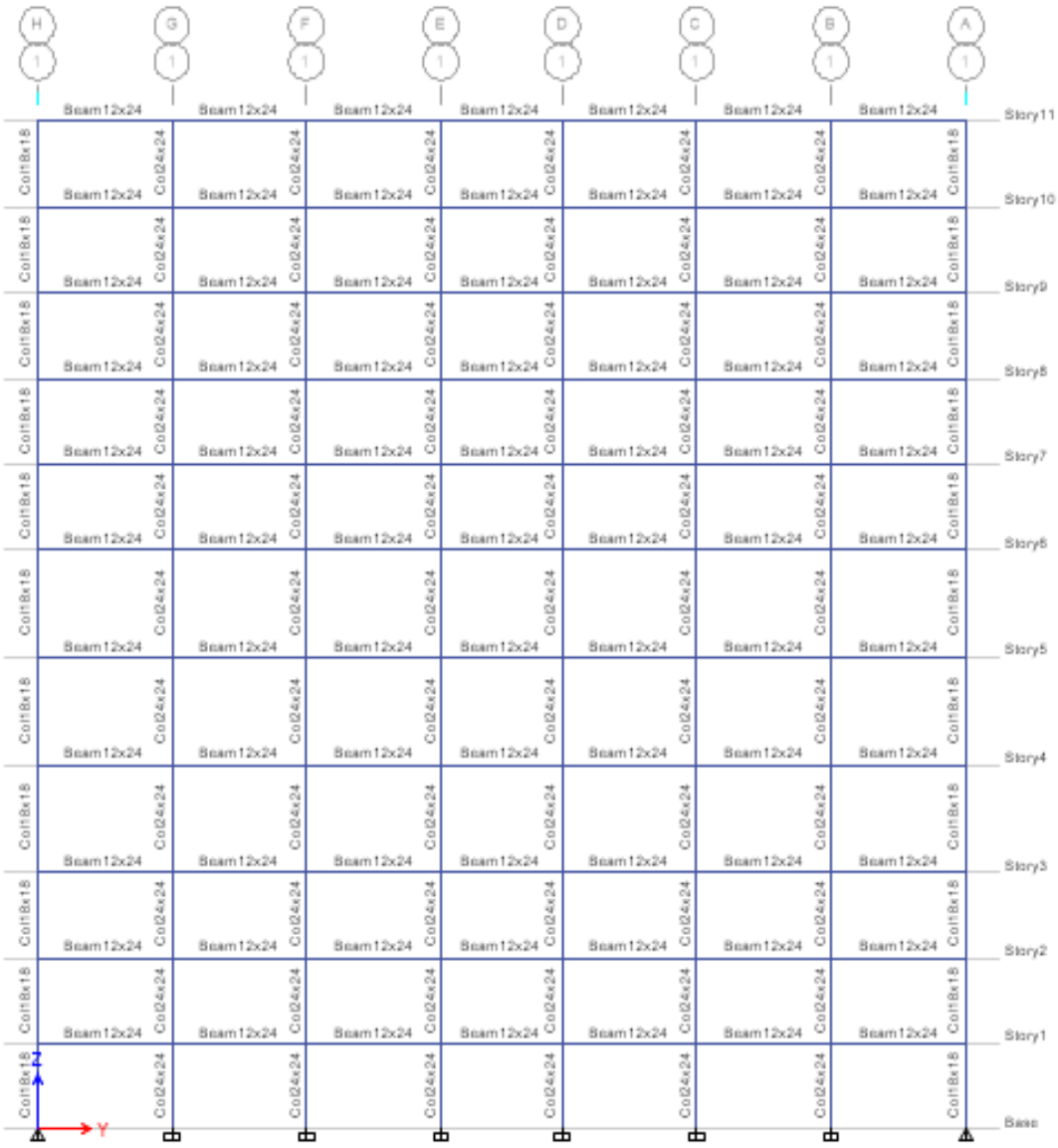
Here, fc' = 4 ksi and fy = 60 ksi (assumed)

Ag = 835.19 sq.inch

Assuming square column, a 28.89" x 28.89" column is obtained.

We are taking 30" x 30" columns.

Columns Layout



Exterior Column Layout



Interior Column Layout

Footings

Determination of thickness of raft foundation

Maximum shear force, Vn= 2320 kip at Column C3(dimension 24*24)

For punching shear, at d/2 distance from column face,

$V_n \leq 4\phi\sqrt{(fc')}*b*d$ where

b= 4*(24+d)

Solving this, d= 44.57”

thickness ,t = (44.57+3+1)”

= 49”

Area= 240*150= 36000 sqft

Shore pile:

Length= 24m = 80 ft(thumb rule)

L/D=50

So, D= 1.6 ft= 19.2 in

= 20 in > 200 mm(ok)

Area= 314 sqin > 650 sqcm(ok)

Perimeter = 2*(240+150)= 1170 ft

Spacing=6”

Number of piles required= 1170*12/26

= 540

Fire Safety:

Structural Fire Resistance

● Slabs: Both the academic and administration buildings incorporate 6” thick slabs constructed with aggregate, offering a 3-hour fire resistance rating as per Table 4.1.1 in Appendix E.

● Columns: The column sizes for these buildings are specified as 18" x 18" and 22" x22". It is noted that with different aggregate materials, these columns can withstand fire for up to 4 hours. We have selected 24"x24" columns so it is on the safer side.

● Masonry Walls: To complement the existing fire resistance achieved by the slabs and columns, clay masonry units have been selected. Specifically, unfilled bricks of clay or shale with a minimum thickness of 5" will be utilized. This choice aligns with the design objective of achieving a minimum 4-hour fire resistance rating for the walls as well.

Water Requirements

Occupancy Categories and Hazards:

● **Market/Mercantile:** Classified as "Ordinary Hazard-1" in Table 4.4.1. Since the Occupancy Category is F-2

বাংলাদেশ গেজেট, অতিরিক্ত, ফেব্রুয়ারি ১১, ২০২১ ২৮৬৭

Table 4.1.1: Fire Resistance of Structures

Concrete Aggregate Type	Minimum Equivalent Thickness of Concrete Walls, Floors, and Roofs for Fire Resistance Rating									
	1 hr		1.5 hr		2 hr		3 hr		4 hr	
	in	mm	in	mm	in	mm	in	mm	In	mm
Siliceous	3.5	89	4.3	109	5.0	127	6.2	157	7.0	178
Carbonate	3.2	81	4.0	102	4.6	117	5.7	145	6.6	168
Sand-light weight	2.7	69	3.3	84	3.8	97	4.6	117	5.4	137
Lightweight	2.5	64	3.1	79	3.6	91	4.4	112	5.1	130

Concrete Aggregate Type	Minimum Column Dimension for Fire Resistance Rating									
	1 hr		1.5 hr		2 hr		3 hr		4 hr	
	In	mm	in	mm	in	mm	in	mm	in	mm
Siliceous	8	203	9	229	10	254	12	305	14	356
Carbonate	8	203	9	229	10	254	11	279	12	305
Sand-light weight	8	203	8.5	216	9	229	10.5	267	12	305

Clay Masonry Unit	Minimum Required Equivalent Thickness of Masonry for Fire Resistance Rating													
	0.5 hr		0.75 hr		1 hr		1.5 hr		2 hr		3 hr		4 hr	
	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm
Brick of clay or shale, unfilled	.7	43	2.0	51	2.3	58	2.85	72	3.4	86	4.3	109	5.0	127
Brick of clay or shale, grouted or filled with perlite, vermiculite, or expanded shale aggregate	2.3	58	2.65	67	3.0	76	3.7	94	4.4	112	5.5	140	6.6	168

	C3	Custodial institution for the incapable adults	1
	C4	Penal and mental institutions for children	1
	C5	Penal and mental institutions for adults	1
D: Healthcare Facilities	D1	Normal medical facilities	2

Table 4.4.1: Fire Protection Flow Requirements*

Building Type**	Sprinkler System (litre/min.)	Standpipe and Hose System (litre/min.)	Duration in Minutes for Building Heights		
			Up to 51 m	51 m to 102 m	Above 102 m
Light hazard- I	1000	1000	30	38	45
Light hazard- II	1900	1900	50	62	75
Ordinary hazard- I	2650	1900	75	95	112
Ordinary hazard-II	3200	1900	75	95	112
Ordinary hazard-III	4800	1900	75	95	112

Buildings	H2	Moderate fire risk storage	3
I: Assembly	I1	Large assembly with fixed seats	1
	I2	Small assembly with fixed seats	1
	I3	Large assembly without fixed seats	1

Notes:

* See also Sec 4.2.2.3.

** Values will be for one riser serving floor area of 1000 m^2.

Light hazard-I	:	Occupancy groups, A1, A2, A3, E1
Light hazard-II	:	Occupancy groups, A4, A5, B, C, D,E2, E3, I2, I4, F1
Ordinary hazard-I	:	Occupancy groups, I1, I3, I5, F2, F3, G1
Ordinary hazard- II	:	Occupancy groups, G2 , H1
Ordinary hazard- III	:	Occupancy groups, H2
Extra hazard	:	Occupancy group J-pressure and flow requirement for this group shall be determined by Fire Department but shall not be less than required value for Ordinary hazard- III

Standpipe and Hose System Requirements

For "Ordinary Hazard-1" building types, Table 4.4.1 recommends a minimum fire protection flow requirement of 1900 liters/minute (l/min). Regardless of the particular occupancy category of each building space, the fire protection system can effectively handle various fire scenarios by choosing the appropriate flow rate of 1900 l/min. In addition to lowering the possibility of property damage, this puts occupant safety first.

Fire Exit

It's evident that fire exits are not part of the project design we currently have. Even if this could be the case for the time being, it's important to realize why fire exits are important to include in future projects. For everyone's protection within the building in the event of an emergency, such as a fire, fire exits are essential. They offer unhindered, uncluttered routes for people to leave swiftly and safely, lowering the possibility of accidents and facilitating the efficient operation of emergency services.

Plumbing:

	Occupancy	Unit of Floor Area in m^2 per Occupant*
F	Mercantile:	
	Retail sales Area, Ground floor	2.3 net
	or Basement	4.6 net
	All other floor	

For Overhead Tank:

The total floor area of the market $= 3358\ m^3$

Occupant load factor $= 4.6$

Occupant No. $= 3358/4.6 = 730$

The volume of water needed in one day (24 hr) $= 30 \times 730 = 21900 = \mathbf{21.9\ m^3}$

For Underground Water Reservoir:

The total ground coverage area of the market $= 5551\ m^3$

Occupant load factor $= 4.6$

Occupant No. $= 5551/4.6 = 1207$

The volume of water needed in one day (24 hr) $= 30 * 1207 = 36210\ L = \mathbf{36.21\ m^3}$

Rainwater Harvesting Consideration:

For Overhead Tank:

Requirement Check:

Plot area of the Project $= \mathbf{11102\ m^2} > \mathbf{300\ m^2}$

So, facilities for rainwater harvesting have been required.

7.4 Rainwater Harvesting Requirements

7.4.1 General

Every building proposed for constructing on plots having extent of 300 sqm or above shall have facilities for conserving and harvesting rainwater.

Total Floor area $= \mathbf{5551\ m^2} > \mathbf{4000\ m^2}$

Own rainwater harvesting system must be on the market.

4.3.5 Rain Water Harvesting System

4.3.5.1 Buildings of total floor area ≥ 4000 m² shall have its own rain water harvesting system as discussed in Chapter 7 Part 8 and installed complying with Section 7.13 Part 8, of this Code. The reservoir capacity shall be a multiple of the area of Ground Coverage of the building and a rain collection coefficient of 0.073.

Preliminary Reservoir Capacity:

Ground Coverage of the building×0.073 = 5551× 0.073 = **406 m²**

[**As per BNBC 2020, Section- 4.3.5.1**]

Table 8.5.1(d): Domestic Water Requirements for Various other Occupancies and Facility Groups

Class of Occupancy	Occupancy Groups	For Full[a] Facilities (*lpcd*)	For Restricted Facilities (*lpcd*)
F: Mercantile	F1: Small Shops and Market	45	30
	F2: Large Shops and Market	45	30
	F3: Refueling Station	70	45

Notes:

[a] For full facility in occupancy classifications A, B, C and D, the water requirement value includes 25% hot water.

RESTRICTED FACILITIES — The minimum plumbing facilities acceptable for the occupants of low income group.

Septic Tank Design:

Table 8.7.1 (Contd.)
Plumbing Fixtures Requirement

Type of Building Occupancy	Water Closets *	Urinals **	Wash Basins ***	Bathtubs or Shower	Drinking Fountains	Other Fixtures
	Add 1 for each additional 150 persons					
Art Galleries, Libraries, Museums (for permanent employee use)	Males: 1 for 1-15 persons 2 for 16-35 persons Females: 1 for 1-12 persons 2 for 13-25 persons	Males: Nil upto 6 persons 1 for 7-20 persons 2 for 21-45 persons	Males: 1 for 1-15 persons 2 for 16-35 persons Females: 1 for 1-12 persons 2 for 13-25 persons	—	1 for 100 persons	—
E5 Sports Facilities	Males: 1 for 75 persons Females: 1 for 50 persons	Males: 1 for 75 persons	1 for 60 persons	1 for 50 persons	1 for 300 persons	Service sink: 1
F Business and Mercantile Building						
F1 Offices F4 Garages and Petrol Stations F5 Essential Services	Males: 1 for 25 persons Female: 1 for 15 persons	Males: Nil upto 6 persons 1 for 7-20 persons 2 for 21-45 persons 3 for 46-70 persons 4 for 71-100 persons Add @ 3% for 101-200 persons and @ 2.5% for over 200 persons.	1 for 25 persons	—	1 for 100 persons	Service sink: 1 per floor.
F2 Small Shops and Markets F3 Large Shops and Markets	1 for 500 persons	Urinals may be provided in toilet room in lieu of water closets for men but for not more than 1/2 of the required number of water closets.	1 for 750 persons	—	1 for 1000 persons	Service sink: 1

(Continued to next page)

Location	Name of the toilets	Male (%)	Male free No. (%)	Female No. (%)	Female Free No. (%)	Reporting Time
Dhaka	Gabtoli Inter Bus Terminal	215634 (65.9)	5715 (1.7)	103337 (31.6)	2315 (0.7)	Uses Jul. 18- Jun. 19
	Farmgate Indira Road	242219 (89.4)	2109 (0.8)	25762 (9.5)	785 (0.3)	
	Shamoli Shisu Park	115038 (85.7)	3140 (2.3)	15196 (11.3)	901 (0.7)	
	Mirpur Zoo Road	105854 (84.0)	1710 (1.4)	18263 (14.5)	183 (0.1)	
	Farmgate Anondho Hall	189922 (91.01)	976 (0.47)	17702 (8.48)	74 (0.04)	
	Mohammadpur	134742 (95.61)	715 (0.51)	5456 (3.87)	20 (0.01)	
	Mirpur 12	43817 (92.06)	735 (1.54)	2762 (5.80)	282 (0.59)	Dec. 18-Jan. 19
Chattogram	Laldighir par	173235 (92.05)	77 (0.04)	14870 (7.90)	21 (0.01)	Jul. 18-Jan. 19
	K.C Day Road	138890 (96.96)	51 (0.04)	4300 (3.00)	-	Jul. 18-Jan. 19
	Bibir Hut	32947 (98.65)	78 (0.23)	374 (1.12)	-	Jan. 19-Oct. 19
	Oxygen Mor	137624 (93.54)	3 (0.00)	9508 (6.46)	-	Jan. 19-Oct. 19
	Natun Bridge	32947 (98.65)	78 (0.23)	374 (1.12)	-	Jan. 19-Oct. 19

Summary of public toilets funded by Water Aid Bangladesh.

Total person use toilet in a day=(134742+715+5456+20)/335=420 (For Floor 1 to Floor 3)

For 1 to 3 floors,

Number of toilets=1+1+1=3

For 4 to 8 Floor,

Number of toilet=5*4=20 (For 71 to 100 persons)

Design of Septic Tank:

Volume required for Sedimentation:

$t_h = 1.5 – 0.3 \log_{10}(Pq)$

> The flow may be computed considering waste water flow 60% to 70% of the water consumption or on the basis of the plumbing fixtures discharging simultaneously into the septic tank, Sec 6.9.12.7. In absence of these data the waste water flow for the Occupancy groups A, C and D may be considered 120 *lpcd* for cities, 50 *lpcd* for district town and 20 *lpcd* for thanas and rural areas. For other Occupancy groups a waste water flow of 10 *lpcd* may be considered.

t_h = minimum hydraulic detention time (days)

P = population served by septic tank

q = wastewater flow (lpcd)

q=120 lpcd

p=420+75*5 =795

t_h=0.63 days

Tank volume (m3) required for sedimentation,

$V_h = 10^{-3}$ P.q.th =60.1 m^3

Volume required for Sludge Digestion:

$t_d = 30 (1.035)^{35-T}$ (Let, T=25˚C)

=42.32 days

$V_d = 0.5 \times 10^{-3}$ P.t_d =16.82 m^3

Volume required for Sludge Storage:

Tank volume (m^3) required for storage of sludge isgiven by,

V_{sl} = C x P x N

C = rate of accumulation of digested sludge

= 0.06 m3/person/yr, for N < 5

= 0.04 m3/person/yr, for N > 5

P = population served by septic tank

N = desludging interval (years)

Let, N=6 years

V_{sl} =0.04*795*6 =190.8 m^3

Volume required for Scum Storage:

Scum accumulation rate = 30 – 40% of sludge accumulation rate

Thus, V_{sc} = 0.40 V_{sl} =76.32 m^3

~~Total Effective Volume of Septic Tank, $V_{eff} = V_{sc} + V_h + V_d + V_{sl}$ =344 m^3~~

• Based on experience, effective depth of a septic tank is usually set between 1.0 to 1.5 m (deeper tanks may be designed, if there is space constraint).

• Higher x-sectional area is advantageous; hence, for the first trial, select an effective depth of about 1.0m, and determine the x-sectional area "A" (round off to nearest reasonable value).

• Then proceed with determination of effective depth of each zone (i.e., Vzone/A). Special consideration is required for sedimentation zone (as explained below).

• A free board of about 250 mm (0.25 m) is usually kept

X-sectional area of septic tank = 344/1.5=230 m^2

Considering rectangular shape, with length =3x width,

Overall internal dimension of tank = 26 m x 9 m

If two compartments are used:

(17 x 9) x 1.5 m + (9 x 9) x 1.5 m

12.3 Preliminary Cost Estimation

Building type: Commercial

Type of structure: R. C. C. frame structure

Soil type: Bearing capacity of soil = 4.32 ksf

Foundation: Deep foundation, 8 storeys and 3 basements

Plinth Area: 36000 sft (3344.5094 sqm)

Basement: Parking

Site: Dhaka (Earthquake zone-II)

Cost Estimation from PWD SCHEDULE OF RATES 2022 (REVISED) PART A - CIVIL WORKS

1. Soil Investigation

		Unit	Amount	Unit Rate (Tk.)	Total Amount (Tk.)
1.	Soil Investigation: LS or Actual cost (BH Nos)	nos	5	38,821.00	194,105.00
				“A” =	194,105.00

2. Foundation Cost

		Unit	Amount	Unit Rate (Tk.)	Total Amount (Tk.)
1.	Foundation Cost	sqm	3346.22	38090.00	127,457,520.00
				“B” =	127,457,520.00

3. Basement Cost

		Unit	Amount	Unit Rate (Tk.)	Total Amount (Tk.)
1.	RCC Retaining Wall	sqm	2610	7,783.00	20313630.00
2.	Basement Floor as Parking	sqm	3346.22	16508.00	55239399.76
3.	Retaining Piles	m	237.8	346,371.00	82367023.80
4.	Bracing Steel	sqm	3346.22	1,755.00	5872616.10
5.	Water proofing: Wall & Foundation Bed	sqm	5956.22	1,890.00	11257255.80
6.	Back Wall to Retaining Wall (125 mm brick work)	sqm	2610	2,077.00	5420970.00
7.	Excavation, carrying & safety	m	12242.27	1,380.00	16894332.60
8.	System management	sqm	3346.22	161.00	538741.42
				“C” =	197,903,969.48

4. Super Structure Cost

			Amount	Unit Rate (Tk)		Total amount (Tk)	
1	i	1st Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	29326	/sqm	98081100.26
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	2069	/sqm	6919791.19

2	i	2nd Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	29766	/sqm	99552684.66
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	1774	/sqm	5933160.74
3	i	3rd Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	30212	/sqm	101044336.1
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	1478	/sqm	4943185.78
4	i	4th Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	30532	/sqm	102114579.3
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	1183	/sqm	3956555.33
5	i	5th Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	30855	/sqm	103194856.1
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	887	/sqm	2966580.37
6	i	6th Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	31026	/sqm	103766767.3
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	592	/sqm	1979949.92
7	i	7th Floor (from PLAR Table-2) 3344.51 sqm	3344.51	sqm	31177	/sqm	104271788.3
	ii	Add extra cost for M.W (from PLAR Table-3) 3344.51 sqm	3344.51	sqm	296	/sqm	989974.96
8	8th Floor (from PLAR Table-2) 3344.51 sqm		3344.51	sqm	31354	/sqm	104863766.5
"D"						=	844,579,076.8

5. Additional Super Structure Cost

1.	Add extra super structure cost 35 for wind (other than coastal area) & earthquake load Resisting structure.	844579076.8		3%	/sqm	25337372.3
2.	Extra cost for Rooftop & LC (From PLAR Table-2) 3344.51 sqm	3344.51	sqm	3629	/sqm	12137226.79
3.	Extra cost for Rooftop R C. C Parapet (From additional cost chart item-3)	237.74	sqm	4902	/sqm	1165401.48
4.	Extra cost for Rooftop R C. C Water Tank(From additional cost chart item-3) 6000 gal.	6000	gal.	196	/sqm	1176000
5.	Extra cost for safety canopy	694.32	sqm	2429	/sqm	1686503.28
6.	Extra cost for safety net	14812	sqm	51	/sqm	755412
"E"					=	42,257,915.85

6. Other Building Cost

Serial No	Name of the work	Quantity	Unit	Unit Rate (Tk)	Total Amount (Tk)
1	Internal Sanitary & Water Supply: Considering 8 floors	2082	sqm	Tk. 1567 per sqm	3,262,494.00
2	Internal Electrification: Considering 8 floors with 3 basement floors	2897	sqm	Tk. 1930 per sqm	5,591,210.00

3	Construction of Underground Reservoir	11001	gallon	Tk. 117 per gallon	1,287,117.00
4	Construction of Overhead Tank	6658	gallon	Tk. 117 per gallon	778,986.00
5.	Construction of nonelectric Eco STP of different sizes (as per detail drawing attached in annexure) with 250 mm walls of brick work in cement mortar (1:4) having a brick flat soling and 125 mm thick reinforced cement concrete flooring (1:2:4) with 125 mm thick walls (1:4) in partition and flush pointing (1:2) on inside wall surface and 12 mm thick cement plaster (1:4) with N.C.F. on floor and slab surface including supplying, fitting and fixing of upvc pipes & Tees and providing 450 mm dia water sealed heavy type C.I. manhole cover with locking/unlocking arrangement and 100 mm thick R.C.C (1:2:4) top slab, including centering, shuttering, fabricating, casting and curing etc. complete up to required depth including necessary earth work in excavation and shoring, bailing out water and side filling, dressing, inside cleaning, cow dung or old septic tank liquid for charging/start up including the cost of all materials, operations and incidental charges etc. all complete as per type plan approved and accepted by the Engineer-in-charge. (Rate is including cost of reinforcement and its fabrication, binding and placing).	1000	each	Tk. 683,652.00 per 500	1,367,304.00

Serial no	**Description**	Unit	Rate (in Tk)	Quantity	Total cost (Tk)
1.	R.C.C. Road: Construction of R.C.C. (minimum f'cr = 30 MPa, and satisfying a specified compressive strength f'c = 25 MPa at 28 days on standard cylinders, crushed stone chips as coarse aggregate, cement content related to mix ratio 1:1.5:3) road with 250mm thick guide wall of height 0.30meter, 150 mm thick R.C.C work over one layer 1st class brick flat soling and polythene sheet including the cost of reinforcement 10 mm dia MS rod 175 mm c/c in both direction	sqm	6,228	4455	27,745,740.00
2.	Boundary wall in RCC Frame, 1.80 m in height, level difference between FGL and EGL up to 0.45m	Meter	15,129	401	6,066,729.00
3.	Barbed wire in fencing over boundary wall work @ 150 mm c/c in both horizontally and vertically.	Meter	1,122	401	449,922.00
				“F”=	46,549,502.00

Electro-Mechanical Component

		Unit	Amount	Unit Rate (Tk.)	Total Amount (Tk.)
1.	Lift (Ordinary)	nos	4	4,200,000.00	16,800,000.00
2	Lift (Service)	nos	1	5,500,000.00	5,500,000.00
				“G” =	23,300,000.00

Fire Safety

		Unit	Amount	Unit Rate (Tk.)	Total Amount (Tk.)
1.	Fire Safety Cost	sqm	26768	968.250	25,918,276.60

				"H" =	25,918,276.60

7. A+B+C+D+E+F +G+H= P	**= 1,308,160,365.73**

8.	Quality Assurance, Material Sample Collection & Testing (1% of P)	"Q"	= 13,081,603.65	1% of "P"
9.	Contingency (5% of P) or Actual need	"R+S"	= 65,408,018.28	5% of "P" or actual need
10.	EMP Cost		= 55.14,000.00	

Grand Total	= **1,386,649,987.66**Tk

Per sft cost = (**1,386,649,987.66/ 36,000*11) = 3500 Tk**

12.4 Revenue for Financial analysis

Total Revenue		
Revenue Source	**BDT (lakh)**	**Rate**
Shop	**343**	**7 lakh per month per 1000sft**
Office	**990**	**5.5 lakh per month per 1000sft**
Food corner	**88.35**	**6 lakh per month per 1000sft**
Basement Storage	**75.6**	**70,000 per 1000sft per month**
Total Revenue	**1,496.95**	

Revenue For Economic analysis

Source	Total Revenue (BDT)
Job opportunity	**71,61,000/-**
Tax Revenue	**11,30,000/-**
Increased property value	**31,55,00,000/-**
Waste and Wastewater management	**12,00,000/-**
Consumer Spending	**36,50,000/-**
Total benefit	**32,86,41,000/-**

Financial Analysis Cash Flow

Year	Undiscounted Cost			Undiscounted Benefits	Net Cash Flow	Internal rate of Return (FIRR)	Benefit Cost Ratio
	Construction Cost (Crore)	Maintenance Cost (Crore)	Total Cost (Crore)	Total Benefit (Crore)	Total Benefit - Total Cost (Crore)		
2025	44.46		44.46		-44.45	28.33	2.70
2026	44.46		44.46		-44.45		
2027	44.46		44.46		-44.45		
2028		4	4	14.96	10.96		
2029		4	4	14.96	10.96		

2030		4	4	14.96	10.96		
2031		4	4	14.96	10.96		
2032		4	4	14.96	10.96		
2033		4	4	17.952	13.952		
2034		4	4	17.952	13.952		
2035		4	4	17.952	13.952		
2036		4	4	17.952	13.952		
2037		4	4	17.952	13.952		
2038		4.6	4.6	21.54	16.94		
2039		4.6	4.6	21.54	16.94		
2040		4.6	4.6	21.54	16.94		
2041		4.6	4.6	21.54	16.94		
2042		4.6	4.6	21.54	16.94		
2043		4.6	4.6	25.85	21.25		
2044		4.6	4.6	25.85	21.25		
2045		4.6	4.6	25.85	21.25		
2046		4.6	4.6	25.85	21.25		
2047		4.6	4.6	25.85	21.25		
2048		5.29	5.29	31.02	25.73		
2049		5.29	5.29	31.02	25.73		
2050		5.29	5.29	31.02	25.73		
2051		5.29	5.29	31.02	25.73		
2052		5.29	5.29	31.02	25.73		
2053		5.29	5.29	37.23	31.94		
2054		5.29	5.29	37.23	31.94		
2055		5.29	5.29	37.23	31.94		

2056		5.29	5.29	37.23	31.94		
2057		5.29	5.29	37.23	31.94		
2058		6.08	6.08	44.67	38.59		
2059		6.08	6.08	44.67	38.59		
2060		6.08	6.08	44.67	38.59		
2061		6.08	6.08	44.67	38.59		
2062		6.08	6.08	44.67	38.59		
2063		6.08	6.08	53.60	47.52		
2064		6.08	6.08	53.60	47.52		
2065		6.08	6.08	53.60	47.52		
2066		6.08	6.08	53.60	47.52		
2067		6.08	6.08	53.60	47.52		
2068		7.00	7.00	64.33	57.33		
2069		7.00	7.00	64.33	57.33		
2070		7.00	7.00	64.33	57.33		
2071		7.00	7.00	64.33	57.33		
2072		7.00	7.00	64.33	57.33		
2073		7.00	7.00	77.19	70.19		
2074		7.00	7.00	77.19	70.19		
2075		7.00	7.00	82.52	75.52		
		Discount Rate	NPV	NPV	NPV		
		0.12	135.84	366.91	231.08		

Economic Analysis Cash Flow

Year	Undiscounted Cost	Undiscounted Benefits	Net Cash Flow	Net Present value	Internal rate of Return

						(crore)	(EIRR)
	Construction Cost (Crore)	**Maintenance Cost (Crore)**	**Total Cost (Crore)**	**Total Benefit (Crore)**	**Total Benefit - Total Cost (Crore)**	**Discount Rate = 6%**	
2024	**35.56**		**35.56**		**-35.56**	130.33	**24%**
2025	**35.56**		**35.56**		**-35.56**		
2026	**35.56**		**35.56**		**-35.56**		
2027		**3.2**	**3.2**	**32.86**	**29.66**		
2028		**3.2**	**3.2**	**32.86**	**29.66**		
2029		**3.2**	**3.2**	**32.86**	**29.66**		
2030		**3.2**	**3.2**	**32.86**	**29.66**		
2031		**3.2**	**3.2**	**32.86**	**29.66**		
2032		**3.2**	**3.2**	**32.86**	**29.66**		
2033		**3.2**	**3.2**	**32.86**	**29.66**		
2034		**3.2**	**3.2**	**32.86**	**29.66**		
2035		**3.2**	**3.2**	**32.86**	**29.66**		
2036		**3.2**	**3.2**	**32.86**	**29.66**		
2037		**3.68**	**3.68**	**37.79**	**34.11**		
2038		**3.68**	**3.68**	**37.79**	**34.11**		
2039		**3.68**	**3.68**	**37.79**	**34.11**		
2040		**3.68**	**3.68**	**37.79**	**34.11**		
2041		**3.68**	**3.68**	**37.79**	**34.11**		
2042		**3.68**	**3.68**	**37.79**	**34.11**		
2043		**3.68**	**3.68**	**37.79**	**34.11**		
2044		**3.68**	**3.68**	**37.79**	**34.11**		
2045		**3.68**	**3.68**	**37.79**	**34.11**		
2046		**3.68**	**3.68**	**37.79**	**34.11**		
2047		**4.23**	**4.23**	**43.46**	**39.23**		
2048		**4.23**	**4.23**	**43.46**	**39.23**		
2049		**4.23**	**4.23**	**43.46**	**39.23**		
2050		**4.23**	**4.23**	**43.46**	**39.23**		
2051		**4.23**	**4.23**	**43.46**	**39.23**		

2052		**4.23**	**4.23**	**43.46**	**39.23**		
2053		**4.23**	**4.23**	**43.46**	**39.23**		
2054		**4.23**	**4.23**	**43.46**	**39.23**		
2055		**4.23**	**4.23**	**43.46**	**39.23**		
2056		**4.23**	**4.23**	**43.46**	**39.23**		
2057		**4.87**	**4.87**	**49.98**	**45.11**		
2058		**4.87**	**4.87**	**49.98**	**45.11**		
2059		**4.87**	**4.87**	**49.98**	**45.11**		
2060		**4.87**	**4.87**	**49.98**	**45.11**		
2061		**4.87**	**4.87**	**49.98**	**45.11**		
2062		**4.87**	**4.87**	**49.98**	**45.11**		
2063		**4.87**	**4.87**	**49.98**	**45.11**		
2064		**4.87**	**4.87**	**49.98**	**45.11**		
2065		**4.87**	**4.87**	**49.98**	**45.11**		
2066		**4.87**	**4.87**	**49.98**	**45.11**		
2067		**5.60**	**5.60**	**57.47**	**51.88**		
2068		**5.60**	**5.60**	**57.47**	**51.88**		
2069		**5.60**	**5.60**	**57.47**	**51.88**		
2070		**5.60**	**5.60**	**57.47**	**51.88**		
2071		**5.60**	**5.60**	**57.47**	**51.88**		
2072		**5.60**	**5.60**	**57.47**	**51.88**		
2073		**5.60**	**5.60**	**57.47**	**51.88**		
2074		**5.60**	**5.60**	**57.47**	**51.88**		